

Replicable Shocks: When Supply Masquerades as Demand*

Giuseppe Faè

Universitat Pompeu Fabra – BSM

Filippo Ferroni

Università di Bologna

Simone Trambaiolo

Università Bocconi

September 10, 2026

Abstract

We formalize shock replicability as the situation where a linear combination of non-target shocks generates the same sign pattern as the shock of interest, and define the associated shock replica. Using NK models as a data-generating process, we quantify the scope that standard sign restrictions leave for replicas, show that near-replicas are reached with positive probability, and trace how they inflate the estimated demand contribution to inflation. Trimming the sign identified IRF with the replica percentiles reduces distortions. We revisit the role of demand disturbances in the 2021–2024 inflation surge.

Keywords: Shock identification and replicability, sign restrictions, forecast error variance decomposition.

JEL classification: C32, E31, E32, E52.

*E-mail: filippo.ferroni2@unibo.it (corresponding author).

1 INTRODUCTION

Stabilization policy hinges on correctly diagnosing the sources of macroeconomic fluctuations. In the New Keynesian paradigm, monetary policy is a powerful tool for managing aggregate demand, whereas direct instruments on the supply side are typically more limited. This asymmetry makes attribution economically consequential. If inflation is primarily demand-driven, tighter monetary policy can restrain spending and ease price pressures. If instead inflation reflects adverse supply forces, compressing demand may lower inflation only at the cost of substantial output losses. Distinguishing between demand and supply sources of inflation is therefore central to policy design.

A large recent literature has studied the 2021–2024 inflation surge using VAR models that distinguish “demand” from “supply” shocks largely through sign restrictions on the joint responses of prices and quantities (e.g., Giannone and Primiceri, 2024; Shapiro, 2024; Bergholt et al., 2024; Ascari et al., 2025). Sign restrictions have become a standard tool in structural VAR analysis because they are flexible, transparent, and often economically plausible. Their appeal is especially strong when exact zero restrictions are hard to justify and the researcher wishes to remain agnostic about parts of the transmission mechanism, as in the case of the recent inflation surge.

Yet the same agnosticism that makes sign restrictions attractive may also make them fragile. If the imposed sign pattern is too permissive, structurally different disturbances may satisfy the same identifying restrictions and therefore remain observationally admissible. A popular example is the monetary policy masquerading discussed in Wolf (2020).¹ However, the issue is more pervasive and applies to many structural shocks. For example, a persistent productivity improvement combined with a temporary cost-push shock can generate the same sign pattern—including positive comovement of output and inflation—that is often interpreted as evidence of a demand disturbance. This paper studies that problem, which we call *shock replicability*. We show that standard demand-supply sign restrictions mechanically favor an interpretation of inflation as demand-driven because supply-driven replicas satisfy the same restrictions. Once those rotations are removed, the demand contribution falls materially.

The paper makes four contributions. First, we develop a general framework for analyzing replicability in linear state-space systems. We define a *shock replica* as a combination of structural shocks that reproduces the target sign pattern while placing zero

¹Wolf (2020, 2022) studies cases in which supply and demand shocks replicate the effects of a monetary policy shock on inflation and the interest rate, and labels this phenomenon *masquerading*. The terminology differs, but the underlying identification problem is closely related.

weight on the target shock, and we introduce two diagnostics: the *replicability mass*, the measure that such a zero-weight combination satisfies the identifying restrictions, and the *replica ratio*, which compares this measure with the measure of an unrestricted combination. Since exact replicas are a measure-zero event, we complement these with the ε -*replica share*: the fraction of the identified set in which the target shock accounts for at most ε^2 of the variance of the identified disturbance. This is the positive-measure counterpart of the replica ratio: it establishes that economically misleading rotations are not merely admissible but are reached with positive probability, and hence that replicability is a property of the identified set rather than of its boundary. Together these objects recast the usual identification question in quantitative terms: how much room do the restrictions leave for misidentification?

Second, we show that replicability is economically relevant in a standard New Keynesian environment, (i) analytically using the three-equation textbook NK model and (ii) taking the Smets and Wouters (2007) model as the data-generating process. In the benchmark models, conventional demand–supply sign schemes allow supply-driven combinations to satisfy the identifying restrictions for a demand disturbance with non-trivial mass. As a result, empirical decompositions based on standard sign restrictions may overstate the role of demand in explaining inflation fluctuations.

Third, we show how to trim impulse responses so that the replicability concerns mitigate. In particular, we show that restrictions on the percentile impact distribution of interest rate, inflation and output responses help distinguish genuine demand shocks from supply-type disturbances that replicate the demand sign pattern. We evaluate these refinements in model-based exercises and quantify the reduction in distortion in forecast error variance decompositions. At the same time, we stress an important caveat: while trimming removes the replica-driven bias, the resulting decompositions remain wide. Even after refinement, the sign restrictions we consider leave the demand contribution only loosely pinned down, so the improvement is best understood as removing a systematic distortion rather than delivering sharp identification.

Fourth, we study these issues in empirical SVARs for the United States and the Euro Area, focusing on the post-2020 inflation episode. Under agnostic sign restrictions, inflation surprises are largely attributed to demand. Under refined identification, the contribution of demand declines, while the decomposition of GDP changes less. The point is that standard agnostic schemes may systematically attribute too much inflation variation to demand because some accepted draws correspond to supply-driven replicas.

Literature review Our framework contributes to the literature on the informativeness of set-identifying restrictions. Baumeister and Hamilton (2015) show that in sign-restricted SVARs the prior over rotations does not vanish asymptotically, so posterior statements about impulse responses or decompositions partly reflect the uniform (Haar) prior conventionally used to explore the identified set. Our diagnostics quantify one specific dimension of this dependence: the replicability mass measures how easily the restrictions are satisfied by combinations that exclude the target shock, while the replica ratio measures how weakly the restrictions penalize such combinations relative to the uniform benchmark. When the replica ratio is zero, replicas are excluded by the restrictions themselves—a statement that is robust to any prior supported on the identified set, in the spirit of the robust Bayesian approach of Giacomini and Kitagawa (2021).

The paper is also related to work that enriches sign restrictions with additional information, including zero restrictions (Arias et al., 2018), narrative restrictions (Antolín-Díaz and Rubio-Ramírez, 2018), elasticities (Kilian and Murphy, 2012), ranking restrictions (Amir-Ahmadi and Drautzburg, 2021), and inequality restrictions on higher-order moments (Andrade et al., 2025), to name a few. Our results rationalize this practice from a new angle: additional restrictions are useful not only because they sharpen the identified set, but also because they can reduce the scope for replica shocks. Our model-based evaluation strategy follows the tradition of Canova and Pautitan (2011), who use DSGE models as laboratories to assess what sign restrictions can recover; relative to their work, we characterize the mass and the location of the misidentifying rotations and design restrictions that target them. The problem we study is distinct from the deformation issues in Canova and Ferroni (2022), where aggregation distortions arise because the data-generating process contains more structural disturbances than the empirical model.

Moreover the paper speaks to the literature identifying the sources of business cycles with a handful of factors, see Angeletos et al. (2020) or Gambetti et al. (2025). Such shocks are typically interpreted in a semi-structural perspective as representing two broad categories (demand and supply) which can include a number of different shocks with similar characteristics in terms of impulse response functions. We caution such interpretation as composite semi-structural demand shocks are not necessarily the result of a combination of individual demand shocks piled up together. We show analytically and with simulations that pure supply disturbances can replicate positive comovement of output and inflation—that is often interpreted as evidence of a demand disturbance.

Finally, our empirical analysis connects to the recent literature on uncertainty in historical decompositions. Bergholt et al. (2024) emphasize that such decompositions may vary widely even across parameter draws associated with similar impulse responses, especially when deterministic components are uncertain. Our paper focuses on a distinct but complementary source of instability: even abstracting from reduced-form estimation uncertainty, the structural rotation itself may be economically misleading because admissible replicas satisfy the same identifying restrictions. Credible historical decompositions therefore require attention both to statistical uncertainty and to the structural informativeness of the identifying scheme.

The rest of the paper proceeds as follows. Section 2 develops the formal framework and introduces the replicability diagnostics. Section 3 illustrates the diagnostics analytically in a three-equation New Keynesian model. Section 4 studies replicability in a New Keynesian benchmark. Section 5 applies the framework to post-pandemic inflation dynamics in the United States and the Euro Area. Section 6 concludes.

2 REPLICABILITY

This section introduces the *replicability* of structural shocks: situations in which shocks other than the one(s) of interest can generate the same sign pattern typically used to identify the target shock (or a set of shocks). We also define a *shock replica*, a combination of structural shocks that generates the intended sign pattern but places zero weight on the target shock. When a shock (or a system of shocks) is fully replicable, a sign-identified SVAR may fail to isolate the intended disturbance, potentially distorting variance and historical decompositions.

2.1 NOTATION

Let $e_j \in \mathbb{R}^n$ denote the j th canonical basis vector (zeros everywhere *except* in position j), e.g. $e_1 = (1, 0, \dots, 0)'$. Define $I_{n,k} = \begin{pmatrix} e_1 & \dots & e_k \end{pmatrix}$, $I_{n,n} \equiv I_n$. Let $\text{sign}(\cdot)$ denote the element-wise sign map taking values in $\{-1, +1\}$, with $\text{sign}(0) = +1$. For an $m \times n$ matrix A , let $\text{vec}(A)$ be the column-wise vectorization of A . Let a_j denote the j th column of A and A_{-j} the matrix obtained by removing the j th column. We denote by $\mathbb{S}^{n-1} = \{x \in \mathbb{R}^n : \|x\| = 1\}$ the unit sphere in $\mathbb{R}^n$.

2.2 STRUCTURAL MODEL

Consider a Gaussian linear state-space representation:

$$\begin{aligned}x_t &= Ax_{t-1} + B\epsilon_t, \\y_t &= Cx_{t-1} + D\epsilon_t,\end{aligned}\tag{1}$$

where x_t is an n_x -dimensional state vector, $y_t \in \mathbb{R}^n$ collects observables, and $\epsilon_t \in \mathbb{R}^n$ is a vector of structural shocks with $\epsilon_t \sim \mathcal{N}(0, I_n)$ i.i.d.

Let F be the $(nh) \times n$ matrix stacking impulse responses of observables up to horizon h :

$$F = \begin{pmatrix} D' & (CB)' & (CAB)' & \dots & (CA^{h-2}B)' \end{pmatrix}'.$$

Each column of F corresponds to a shock; each row corresponds to a particular variable at a particular horizon.

Let S be a $(J \times nh)$ selection matrix (with entries in $\{0, 1\}$) picking the responses used for identification, and define $\tilde{F} \equiv SF$. Assume we are interested in the first k shocks, $\epsilon_{1,t}, \dots, \epsilon_{k,t}$ with $k < n$. Define the target sign pattern

$$v \equiv \text{vec}\left(\text{sign}\left(\tilde{F}I_{n,k}\right)\right) = \text{vec}\left(\text{sign}\begin{pmatrix} \tilde{f}_1 & \dots & \tilde{f}_k \end{pmatrix}\right) \in \{-1, +1\}^{Jk}.\tag{2}$$

No-self-use combinations. To study whether the sign pattern of the target shocks can be replicated *without using the shocks themselves*, consider matrices $Q \in \mathbb{R}^{n \times k}$ satisfying

$$Q'Q = I_k, \quad e_j'q_j = 0 \quad \text{for } j = 1, \dots, k,$$

where q_j is the j th column of Q . Q is a tall rectangular orthonormal matrix with zeros on the main diagonal. The implied sign pattern from these no-self-use combinations is

$$v_Q \equiv \text{vec}\left(\text{sign}\left(\tilde{F}Q\right)\right) \in \{-1, +1\}^{Jk}.\tag{3}$$

Definition 1 (Replicability) *For a given target sign pattern v , the system of shocks $\epsilon_{1,t}, \dots, \epsilon_{k,t}$ is replicable if there exists a matrix $Q \in \mathbb{R}^{n \times k}$ such that (i) $Q'Q = I_k$, (ii) $e_j'q_j = 0$ for all $j = 1, \dots, k$, and (iii) $v_Q = v$, where v and v_Q are defined in (2) and (3) respectively.*

Mechanically, replicability becomes easier as n grows (more shocks to combine) and harder as J grows (more sign restrictions to match). Replicability also becomes harder as k grows: the no-self-use constraints require each q_j to be orthogonal to e_j while the

columns of Q must remain mutually orthonormal, and these two requirements interact.

Definition 2 (Shock replica) *Suppose the target shock $\epsilon_{j,t}$ is replicable. Any $q \in \mathbb{S}^{n-1}$ with $e'_j q = 0$ that satisfies $\text{sign}(\tilde{F}q) = \text{sign}(\tilde{f}_j)$ defines a replica shock $\epsilon_{j,t}^r \equiv q' \epsilon_t$.*

A replica shock satisfies the identifying sign restrictions but assigns zero weight to the true target shock; it therefore represents a worst-case observationally admissible disturbance under the imposed restrictions.

2.3 SVAR AND TRUE DISTURBANCES

Under standard conditions (e.g. Fernández-Villaverde et al., 2007), the observables admit a VAR representation:

$$y_t = \Phi(L)y_{t-1} + u_t, \quad (4)$$

where u_t is i.i.d. with covariance matrix Σ . Assume large samples so the reduced-form VAR and its impulse responses are known, and coincide with the structural model responses up to an orthogonal rotation. A structural VAR (SVAR) representation can be written as

$$u_t = \Sigma^{1/2} \Omega \nu_t, \quad \Omega' \Omega = I_n, \quad \nu_t \sim \mathcal{N}(0, I_n),$$

and there is a continuum of different SVAR consistent with the second moments of the observed data. Let Ψ stack reduced-form impulse responses up to horizon h :

$$\Psi = \left((\Sigma^{1/2})' \quad (\Psi_1 \Sigma^{1/2})' \quad \dots \quad (\Psi_{h-1} \Sigma^{1/2})' \right)'.$$

The SVAR-implied sign pattern for the first k shocks is

$$w \equiv \text{vec} \left(\text{sign} \left(\tilde{\Psi} \Omega I_{n,k} \right) \right), \quad \tilde{\Psi} \equiv S \Psi.$$

Sign identification seeks rotations Ω such that $w = v$. If the target system is replicable, there exist admissible rotations that satisfy the sign restrictions while placing zero weight on the true target shocks, so sign restrictions alone cannot guarantee that the identified shocks correspond to the intended disturbances.

A well-known example is the masquerading (replicability) of monetary policy shocks by combinations of supply and demand shocks in New Keynesian models (see Wolf, 2020, 2022, and Appendix D). In this paper, we study the replicability of demand shocks.

The practical question is how likely this is: for a given structural model and a given set of sign restrictions, (i) how large is the set of no-self-use combinations that replicate the target sign pattern in absolute terms, and (ii) relative to all the admissible combinations and (iii) what are the properties of the resulting replica shocks. We address these questions next.

2.4 CHARACTERIZING THE REPLICABILITY SET

We start with the case of a single target identified shock ($k = 1$) and then consider multiple-shock restriction systems.

Single-shock replicability ($k = 1$). Assume the target is $\epsilon_{1,t}$. The sign pattern is $v = \text{sign}(\tilde{f}_1) \in \{-1, +1\}^J$. For $q \in \mathbb{S}^{n-1}$ with $e'_1 q = 0$, define $v_q \equiv \text{sign}(\tilde{F}q)$. We want the set of q such that $v_q = v$. Matching signs is equivalent to the inequalities

$$Mq > 0, \quad \|q\| = 1, \quad e'_1 q = 0,$$

where $M \equiv \text{diag}(\text{sign}(\tilde{f}_1)) \tilde{F}$. Define the set of sign restrictions holding with $\mathcal{K} \equiv \{q \in \mathbb{R}^n : Mq > 0\}$, and the set of no-self-use combinations with $\mathcal{H} \equiv \{q \in \mathbb{R}^n : e'_1 q = 0\}$. The feasible set is

$$\mathcal{F} \equiv \mathbb{S}^{n-1} \cap \mathcal{K} \cap \mathcal{H}.$$

When $n = 2$ $\mathcal{F}$ is either empty or consists of the two points $(0, \pm 1)'$. When $n = 3$, $\mathcal{F}$ is either empty or contains an open arc on the unit circle, hence a continuum of solutions. See Section 3 for an analytical example. Given $q \in \mathcal{F}$, the corresponding replica shock is

$$\epsilon_{1,t}^r = q' \epsilon_t.$$

When $\mathcal{F}$ is nonempty, sign-restricted SVARs cannot distinguish $\epsilon_{1,t}$ from $\epsilon_{1,t}^r$, i.e. the worst-case observationally admissible disturbance under the imposed restrictions.

Replicability mass and replica ratio. Two scalar measures summarize the size of $\mathcal{F}$. Both are defined with respect to the uniform (Haar) distribution, the reference measure conventionally used to explore the identified set of sign-restricted SVARs (Rubio-Ramírez et al., 2010). The *replicability mass* is the probability that a combination placing no weight on the target shock satisfies the identifying restrictions,

$$\delta \equiv \Pr(Mq > 0 \mid e'_1 q = 0, \|q\| = 1),$$

where q is drawn uniformly on the great subsphere $\{q \in \mathbb{S}^{n-1} : e_1' q = 0\}$, a copy of $\mathbb{S}^{n-2}$ embedded in $\mathbb{R}^n$. The *admissibility mass* is the probability that an unrestricted combination satisfies the restrictions,

$$\mu \equiv \Pr(Mq > 0 \mid \|q\| = 1),$$

where q is drawn uniformly on $\mathbb{S}^{n-1}$. The *replica ratio* compares the two:

$$\gamma \equiv \frac{\delta}{\mu}.$$

The replica ratio measures how easily the target sign pattern can be matched *without* the target shock, relative to how easily it can be matched at all. When $\gamma = 0$, the sign pattern cannot be reproduced by zero-weight combinations: the restrictions are unforgeable. When γ is close to (or above) one, excluding the target shock hardly reduces the probability of satisfying the restrictions: the identifying scheme carries little or no information about the weight that the identified rotation places on the shock of interest.

Remark 3 (Exact replicas are a measure-zero event) *Since $\{q : e_1' q = 0\}$ is a lower-dimensional subset of the sphere, the probability that a draw from the admissible set is an exact replica is zero, and a definition of γ as the conditional probability $\Pr(e_1' q = 0 \mid Mq > 0)$ would be degenerate. The ratio γ has instead a precise interpretation in terms of near-replicas. For $\varepsilon > 0$, call an admissible q an ε -replica if $q_1^2 \leq \varepsilon^2$, i.e. if the target shock accounts for at most ε^2 of the variance of the identified shock $q' \epsilon_t$. Then, as $\varepsilon \rightarrow 0$,*

$$\Pr(q_1^2 \leq \varepsilon^2 \mid Mq > 0) = \gamma \cdot \Pr(q_1^2 \leq \varepsilon^2) \cdot (1 + o(1)),$$

see next section for a detailed discussion. That is, γ is the factor by which the identifying restrictions discount ($\gamma < 1$) or favor ($\gamma > 1$) rotations that nearly exclude the target shock, relative to the uniform benchmark. Being a ratio of masses, γ is a likelihood ratio rather than a probability and can in principle exceed one.

Systems of shocks. The definitions extend naturally to settings where sign restrictions are imposed on several shocks at once, whether the target remains a single shock (e.g. a demand shock identified jointly with a supply shock, as in Sections 4 and 5) or the object of interest is the full system of restricted shocks. Each restricted shock contributes its own set of inequalities, the columns of the rotation matrix must

be mutually orthonormal, and the zero-weight constraint applies to the target column or, for system replicability, to every column, so that no shock loads on itself. Replicability mass, admissibility mass, and replica ratio are defined exactly as above, as conditional masses under the uniform (Haar) distribution on the set of orthonormal matrices. Appendix C provides the formal definitions and the algorithm used to sample the corresponding conditional measures.

Discussion A few remarks are useful. First, these measures are defined relative to a reference structural model whose shocks of interest are invertible, e.g. see Wolf (2020) or Fernández-Villaverde et al. (2007).

Second, researchers are often interested in *combinations* of structural shocks rather than individual disturbances. For instance, a model may feature multiple demand shocks (e.g., preference and investment-demand) and multiple supply shocks (e.g., markup, technology, oil). One could define category-level replicability, where the replica is constrained to be orthogonal to the entire subspace of demand disturbances. Or one could consider a synthetic combination of category level, e.g. demand, shocks and study its replicability.² We discuss such an example in Section 4.5.

Third, these diagnostics also speak to the prior sensitivity of set-identified inference. Since the rotation prior does not update with the data (Baumeister and Hamilton, 2015), a large γ implies that posterior averages over the identified set are heavily influenced by rotations that contain (almost) none of the target shock. Conversely, a zero γ implies that replicas are ruled out by the restrictions themselves, a conclusion that holds for any prior supported on the admissible set (Giacomini and Kitagawa, 2021), see also next section.

Finally, the framework naturally extends beyond sign restrictions. One can incorporate magnitude restrictions, elasticities, or weighted restrictions (placing greater weight on certain horizons or variables) by modifying the inequality system accordingly.

2.5 THE TARGET LOADING: NEAR-REPLICAS AND ROBUST BOUNDS

The diagnostics δ , μ and γ are *masses*: they measure how much of the reference measure sits in the replica region. A complementary and equally natural question is a *magnitude*: how much of the target shock does an admissible rotation actually

²Let $a \in \mathbb{R}^n$ be a vector of weights defining a composite shock $\eta_t \equiv a'\epsilon_t$. Its impulse responses are given by Fa , i.e. a weighted linear combination of the columns of F . Therefore, the same replicability concepts and measures developed above apply directly to composite objects by replacing the target column $\tilde{f}_j$ with the target composite response Fa (and, for systems, replacing $I_{n,k}$ with an appropriate set of weight vectors).

contain? This subsection introduces that object, clarifies its relationship with γ , and shows that the two answer different questions—one about measure, the other about support. Throughout, write

$$\mathcal{G} \equiv \mathbb{S}^{n-1} \cap \mathcal{K}$$

for the identified (admissible) set, so that the replica set defined above is $\mathcal{F} = \mathcal{G} \cap \mathcal{H}$.

Definition 4 (Target loading) For $q \in \mathbb{S}^{n-1}$, define the target loading

$$\tau(q) \equiv (e_1' q)^2 = q_1^2.$$

Because $\epsilon_t \sim \mathcal{N}(0, I_n)$ and $\|q\| = 1$, the identified shock $q' \epsilon_t$ has unit variance and $\tau(q)$ is exactly the share of that variance contributed by the true target shock $\epsilon_{1,t}$. Thus $\tau(q) = 1$ if and only if the identified shock *is* the target shock, and $\tau(q) = 0$ if and only if it is a replica: $\mathcal{F} = \{q \in \mathcal{G} : \tau(q) = 0\}$. The loading is the most direct measure of how “truthfull” an admissible rotation is, and it is immediate to compute for every accepted draw.

The unrestricted benchmark. The distribution of τ across accepted rotations cannot be read in isolation, because τ is already small—and mechanically smaller as n grows—*before* any restriction is imposed. If q is drawn uniformly on $\mathbb{S}^{n-1}$, then³

$$\tau(q) \sim \text{Beta}\left(\frac{1}{2}, \frac{n-1}{2}\right), \quad \mathbb{E}[\tau(q)] = \frac{1}{n}, \quad (5)$$

and, expanding this distribution function at the origin,

$$\Pr(\tau(q) \leq \varepsilon^2) = 2c_n \varepsilon + O(\varepsilon^3), \quad c_n \equiv \frac{\Gamma(n/2)}{\sqrt{\pi} \Gamma((n-1)/2)} \sim \sqrt{n/(2\pi)} \text{ as } n \rightarrow \infty, \quad (6)$$

where c_n is the reciprocal of the Beta normalizing constant.⁴ Two implications are worth stating. First, a purely random rotation already assigns only $1/n$ of the identified shock’s variance to any given structural disturbance: $1/3$ with three shocks, $1/7$ with seven. Second, by (6) the unrestricted frequency of near-replicas *grows* like $\sqrt{n}$: with $\varepsilon = 0.1$, an unrestricted rotation is an ε -replica with probability 0.10 when $n = 3$ and 0.19 when $n = 7$. Near-replicas are not rare events that restrictions must create; they are common events that restrictions must *remove*.

³See Baumeister and Hamilton (2015).

⁴The Beta(1/2, ·) density has a $t^{-1/2}$ singularity at the origin, so the interval $[0, \varepsilon^2]$ carries mass of order ε rather than ε^2 : $\int_0^{\varepsilon^2} t^{-1/2} (1-t)^{(n-3)/2} dt = 2\varepsilon + O(\varepsilon^3)$. This linear rate is what makes the limit in Proposition 5 finite and non-zero.

The replica ratio as a near-replica density ratio. Proposition 5 restates Remark 3 and makes the connection between γ and the loading precise.

Proposition 5 *Suppose that boundary of $\mathcal{K}$, i.e. $\partial\mathcal{K} \equiv cl(\mathcal{K})/int(\mathcal{K})$, intersects the great subsphere $\mathcal{H} \cap \mathbb{S}^{n-1}$ in a set of zero $(n-2)$ -dimensional measure. Then*

$$\lim_{\varepsilon \rightarrow 0} \pi_{\tau}(\varepsilon) = \lim_{\varepsilon \rightarrow 0} \frac{\Pr(\tau(q) \leq \varepsilon^2 \mid q \in \mathcal{G})}{\Pr(\tau(q) \leq \varepsilon^2)} = \frac{\delta}{\mu} = \gamma.$$

The proof is in appendix B. Proposition 5 says that γ is not a competing object to the loading: it is a specific *functional* of the distribution of τ , namely its density at the origin normalized by the unrestricted density at the origin. Being a ratio of densities rather than a probability, γ can exceed one.

Why the normalization matters. The normalization in Proposition 5 is what makes the diagnostic portable, and it removes two distinct confounds. The first is dimensionality. Suppose one reports that the median admissible rotation has $\tau = 0.14$. With $n = 7$ this is exactly the unrestricted benchmark (5), so the correct reading is that the restrictions are doing nothing; with $n = 3$, where the benchmark is 0.33, the same number means that the restrictions are actively *pushing* the identified shock toward the replica region. The identical statistic supports opposite conclusions, and only n distinguishes them. The second confound is the overall tightness of the scheme. A very aggressive set of restrictions that rejects most rotations will mechanically deliver a small δ , whether or not it discriminates against replicas in particular; dividing by μ asks whether the scheme is *selectively* hard on zero-weight combinations rather than uniformly hard on everything. Because both confounds are netted out, γ is comparable across models, across dimensions, and across identifying schemes, while δ and the raw distribution of τ are not.

Robust bounds on the loading. All of δ , μ , γ , and every quantile of τ over $\mathcal{G}$ are statements under the uniform reference measure. Since the rotation prior is never revised by the data (Baumeister and Hamilton, 2015), a different prior supported on $\mathcal{G}$ would deliver different values and no amount of data could adjudicate. The features of the problem that survive this critique are those of the identified set itself. Define the identified set for the loading,

$$\underline{\tau} \equiv \inf_{q \in \mathcal{G}} \tau(q), \quad \bar{\tau} \equiv \sup_{q \in \mathcal{G}} \tau(q). \quad (7)$$

Every prior supported on $\mathcal{G}$ assigns probability one to $\mathcal{G}$, hence probability one to $\{\tau \geq \underline{\tau}\}$: the bound $\underline{\tau}$ holds under *all* such priors, and $[\underline{\tau}, \bar{\tau}]$ is the identified set for the target loading in the sense of Giacomini and Kitagawa (2021). The interpretation is direct and prior-free: under the maintained restrictions, no admissible rotation can be less than $100\underline{\tau}$ percent genuine.

Proposition 6 *Let $\mathcal{F} = \mathcal{G} \cap \mathcal{H}$ denote the replica set. Then*

- (i) $\delta = 0 \iff \mathcal{F} = \emptyset \iff \gamma = 0$;
- (ii) $\delta > 0 \implies \underline{\tau} = 0$, and the infimum is attained;
- (iii) $\underline{\tau} > 0 \implies \delta = \gamma = 0$, but the converse fails: $\gamma = 0$ and $\underline{\tau} = 0$ can hold simultaneously, precisely when $\text{cl}(\mathcal{G}) \cap \mathcal{H} \neq \emptyset$ while $\mathcal{G} \cap \mathcal{H} = \emptyset$.

The proof is in appendix B. Part (ii) is the implication that binds in practice, and it is worth stating in words. *Whenever the identifying restrictions admit replicas at all, the robust lower bound is degenerate*: there exist admissible rotations containing no target shock whatsoever, so no positive floor on the loading can be guaranteed. Because this is a statement about the support of the identified set rather than about its measure, it holds under *every* prior supported on $\mathcal{G}$, and is in that sense a sharper rendering of the identification failure than any value of γ . It also implies that $\underline{\tau}$ is uninformative as a diagnostic on the raw sign-identified set: $\delta > 0$ in every application we consider in Sections 4 and 5, so $\underline{\tau} = 0$ throughout. The bound becomes informative only on refined sets, a point we return to below.

Part (iii) records the one configuration in which measure and support diverge. When it binds, the restrictions exclude every *exact* replica—so $\gamma = 0$ and the scheme is “unforgeable”—and yet the admissible set still contains rotations arbitrarily close to pure replicas. By Proposition 5 these ε -replicas are vanishingly rare relative to the uniform benchmark and are therefore invisible to any measure-based diagnostic; they are nonetheless fully visible in the reported output, because the bounds of a sign-identified set are determined by its extreme draws rather than its typical ones (Fry and Pagan, 2011). In short, γ rules out replicas *in measure*, $\underline{\tau}$ rules them out *in support*, and it is the latter that governs the width of the reported identified set. The two configurations are not merely theoretical possibilities: both arise in the three-equation model of Section 3.

Finite- ε diagnostics. Proposition 5 defines γ as a limit, and exact replicas are a measure-zero event. Both facts invite the objection that these diagnostics describe a

knife-edge rather than the identified set a researcher actually works with. The remedy is to evaluate the same object at a fixed and economically interpretable tolerance. Define the *finite- ε replica ratio*

$$\pi_\tau(\varepsilon) \equiv \frac{\Pr(\tau(q) \leq \varepsilon^2 \mid q \in \mathcal{G})}{\Pr(\tau(q) \leq \varepsilon^2)}, \quad (8)$$

so that $\pi_\tau(\varepsilon) \rightarrow \gamma$ as $\varepsilon \rightarrow 0$ by Proposition 5. We set $\varepsilon = 0.1$ throughout, so that an ε -replica is an admissible rotation in which the target shock accounts for at most one percent of the variance of the identified disturbance.

Two quantities follow from (8), and we report both. The ratio itself says whether the limiting diagnostic survives at a finite tolerance: when $\pi_\tau(\varepsilon) \approx \gamma$, the restrictions favor near-replicas by the same factor at $\varepsilon = 0.1$ as in the limit, so γ is not an artifact of the limiting argument. The implied *share* of the identified set occupied by near-replicas,

$$\Pr(\tau(q) \leq \varepsilon^2 \mid q \in \mathcal{G}) = \pi_\tau(\varepsilon) \cdot \Pr(\tau(q) \leq \varepsilon^2) \approx 2c_n \varepsilon \pi_\tau(\varepsilon), \quad (9)$$

is a probability carrying strictly positive measure. Statements about $\mathcal{F}$, or about the bounds of the identified set, establish only that misleading rotations are admissible; (9) establishes that they are *reached with positive probability*, which is a property of the identified set rather than of its boundary.

Two cautions attach to this statistic and we state them explicitly, since both are easily turned against the argument. First, the benchmark against which the share must be read is itself non-trivial: under the uniform measure with no restrictions imposed, a fraction $\Pr(\tau \leq \varepsilon^2) \approx 2c_n \varepsilon$ of rotations are already ε -replicas—exactly ten percent when $n = 3$ and $\varepsilon = 0.1$ —so a restricted share below that value means the restrictions do discriminate against near-replicas, by precisely the factor $\pi_\tau(\varepsilon)$. Second, (9) is a tail statistic. It shows that the replica region is attainable; it does not measure how far a posterior summary is displaced, which depends on the whole distribution of τ over $\mathcal{G}$ rather than on its lower tail. We therefore use the share to establish that replicability is not a measure-zero curiosity, and turn to the distribution of τ , and to direct measurement of the decomposition error, for the magnitude of the distortion.

3 ANALYTICAL EXAMPLE

In this section we consider a standard three-equation NK model, see Gali (2008). We are interested in identifying the demand shock, and –consistently with the model– postulate that a demand shock generates a comovement between output and inflation.

We then compute analytically its *replicability mass* δ , the probability that a combination of the two supply shocks alone reproduces the target sign pattern, and the demand *replica ratio* γ , which compares this probability with the probability that an unrestricted combination of the model's shocks does. The key equations of the model are:

$$\pi_t = \beta E_t \pi_{t+1} + \lambda(\sigma + \varphi)y_t - \lambda(1 + \varphi)a_t + \sigma_s \epsilon_t^s \quad \text{NKPC} \quad (10)$$

$$y_t = E_t y_{t+1} - \frac{1}{\sigma}(i_t - E_t \pi_{t+1} - \bar{\rho}) + \sigma_d \epsilon_t^d \quad \text{DIS} \quad (11)$$

$$i_t = \bar{\rho} + \phi_\pi E_t \pi_{t+1} \quad \text{MP rule} \quad (12)$$

$$a_t = \rho a_{t-1} + \sigma_a \epsilon_t^a \quad \text{technology} \quad (13)$$

The first equation is the New Keynesian Philipps curve (NKPC) where $\lambda = \frac{(1-\theta)(1-\beta\theta)}{\theta}$ represents the slope of the NKPC, and $1 - \theta$ and β denote the Calvo probability of price adjustment and β is the time discount factor (with $\bar{\rho} = \ln 1/\beta$), respectively. The NKPC is shocked by a cost-push or markup shock, $\epsilon_t^s \sim N(0, 1)$.⁵ Relative to textbook version in terms of 'gaps', we express the NKPC as a function of the marginal cost which is a function of output and technology.⁶ This version allows us to have neater restrictions on inflation and output which will be used for the identification of demand shocks. The second equation is the dynamic IS curve where $\epsilon_t^d \sim N(0, 1)$ is a demand shock. The demand shock can be thought as a risk-premium disturbance capturing fluctuations in households' intertemporal preferences.⁷ The third equation is the monetary policy rule where the interest rate responds to expected inflation; we assume that the Taylor principle is satisfied, i.e. $\phi_\pi > 1$. The last equation is the law of motion of technology where $\epsilon_t^a \sim N(0, 1)$. σ_j for $j = s, d, a$ are standard deviations of price markup, demand and technology shock, respectively. The remaining parameters have the usual interpretation.

Inflation and output jump on impact after a demand shock; but a purely transitory demand disturbance is invisible to a forward looking rule, so interest rate does not react. A positive technology shock reduces the marginal cost persistently and thus decreases inflation and increases output similarly. Since this effect changes inflation expectations, also the interest rate reacts and declines. Finally, the cost push shock increases inflation on impact. To identify demand shocks in the SVAR, we postulate the following restriction:

⁵See Smets and Wouters (2007) for micro foundation of this shock.

⁶To ease the notation and the algebra, we assume a constant-returns-to-scale production function.

⁷Fisher (2015) show that the risk-premium shock in SW can be interpreted as a structural shock to the demand for safe and liquid assets such as short-term U.S. Treasury securities.

Assumption 1 *A demand shock moves prices and quantities in the same direction on impact, i.e. $\pi_t > 0$ and $y_t > 0$.*

The following proposition characterizes the measures of interest.

Proposition 7 *Consider the NK model in (10)–(13). Any SVAR demand shock is a unit-length combination of the model's shocks, $\nu_t^d = q' \epsilon_t$ with $q = (q_1 \ q_2 \ q_3)' \in \mathbb{S}^2$, and Assumption 1 is equivalent to $m_1 q > 0$ and $m_2 q > 0$, where*

$$m_1 = \begin{pmatrix} \lambda(\varphi + \sigma)\sigma_d & -\omega\sigma_a & \sigma_s \end{pmatrix}, \quad m_2 = \begin{pmatrix} \sigma_d & \eta\sigma_a & 0 \end{pmatrix},$$

$\chi = 1 - \beta\rho + \lambda(\sigma + \varphi)(\phi_\pi - 1)\rho/(\sigma(1 - \rho)) > 0$, $\omega = \lambda(1 + \varphi)/\chi > 0$ and $\eta = (\phi_\pi - 1)\rho\omega/(\sigma(1 - \rho)) > 0$. The replicability mass of the demand shock and the demand replica ratio are given by

$$\delta = \frac{1}{2\pi} \arctan\left(\frac{\sigma_s}{\omega\sigma_a}\right)$$

$$\gamma = \frac{\delta}{\mu} = \frac{\arctan\left(\frac{\sigma_s}{\omega\sigma_a}\right)}{\pi - \alpha},$$

where $\delta = \Pr(m_1 q > 0, m_2 q > 0 \mid q_1 = 0)$, $\mu = \Pr(m_1 q > 0, m_2 q > 0) = \frac{\pi - \alpha}{2\pi}$, and $\alpha = \arccos\left(\frac{m_1' m_2}{\|m_1\| \|m_2\|}\right)$ is the angle between the two restrictions.

The proof is in the appendix B. Following Remark 3, γ measures how strongly the demand restrictions discriminate against combinations that contain (almost) no demand shock: values close to one indicate that supply-only combinations satisfy the demand restrictions (almost) as easily as unrestricted combinations. The facility of replicating the demand shock depends on the markup shock variance. When $\alpha > 0$, as the variance of the markup shock increases (relative to that of technology), the replica ratio increases, since the numerator increases and the denominator decreases. The latter is due to the fact that the angle between the two restrictions changes when markup shocks are larger. The dot product does not change but the norm of m_1 increases; this implies that α increases since $\arccos(\cdot)$ is a monotonically decreasing function of its argument. Therefore, $\pi - \alpha$ decreases leading to an increase γ .

Remark 8 (Attainment and tangency) *For $\sigma_s > 0$, $\delta > 0$ and part (ii) of proposition 6 gives $\underline{\tau} = 0$, attained. For $\sigma_s = 0$, instead, imposing $q_1 = 0$ requires simultaneously $-\omega\sigma_a q_2 > 0$ and $\eta\sigma_a q_2 > 0$, which is impossible: $\mathcal{F} = \emptyset$ and $\delta = 0$. Yet setting $q_1 = \varepsilon > 0$ the two inequalities read*

$$-\frac{\sigma_d}{\eta\sigma_a} \varepsilon < q_2 < \frac{\lambda(\varphi + \sigma)\sigma_d}{\omega\sigma_a} \varepsilon,$$

a nonempty interval for every $\varepsilon > 0$. Admissible rotations therefore exist arbitrarily close to the replica hyperplane, and $\underline{\tau} = 0$ without being attained. The knife-edge case $\sigma_s = 0$ is exactly the tangency of part (iii) of proposition 6.

Interestingly, in the extreme case where the true DGP does not feature demand shocks, Assumption 1 still implies a non-trivial mass of demand shock in the SVAR model. When $\sigma_d = 0$, the angle between the two constraints, α , simplifies to $\alpha = \arccos\left(-\frac{\omega\sigma_a}{\sqrt{\omega^2\sigma_a^2 + \sigma_s^2}}\right)$. Therefore, probability of finding an admissible rotations is⁸ $\mu = \Pr(q \in \mathcal{G}) = \frac{\arctan(\sigma_s/(\omega\sigma_a))}{2\pi}$, which is not zero and coincides with the replicability mass δ . In this extreme case $\gamma = 1$: when the data-generating process contains no demand shock, the combinations satisfying the demand restrictions are, in the limit, exclusively supply-driven replicas.

Figure 1a reports the admissibility mass (μ) and the demand replica ratio (γ) as a function of the standard deviation of the markup shock and the demand shock; remaining parameters have standard calibrated values, i.e. $\beta = 0.99$, $\sigma = 1$, $\varphi = 1$, $\theta = 0.75$, $\phi_\pi = 1.5$, $\rho = 0.9$ and $\sigma_a = 1$. The left panel of Figure 1a reports the admissibility mass μ with a light blue line and the replica ratio γ with a dark blue line as a function of the standard deviation of the markup shock (we assume that $\sigma_d = \sigma_a = 1$). While the admissibility mass barely changes when markup shock volatility increases, the replica ratio jumps quickly from zero to about 0.75 when the standard deviation of the markup shock is half of that of the other supply shock and it approaches one when its volatility is double the size of that of technology. The right panel of Figure 1a reports analogous quantities when the standard deviation of the demand shock varies from zero to two, assuming $\sigma_s = \sigma_a = 1$. It is interesting to notice that, when demand shocks are muted, the admissibility mass is non zero, about 0.2. Moreover, the replica ratio declines as the demand shock volatility rises but does not vanish, reaching 0.78 when the demand shock standard deviation is double the size of that of supply shocks.

A large replica mass matters only if replicas carry different economic content from the true demand shock. Panels (b) and (c) of Figure 1 show that they do. Panel (b) reports the impact responses of inflation and output to the sign-identified demand shock and to the replica, as a function of the markup volatility σ_s . The true demand shock moves inflation and output by comparable amounts; the replica, being a combination of supply shocks, generates a much stronger inflation response relative to output, and

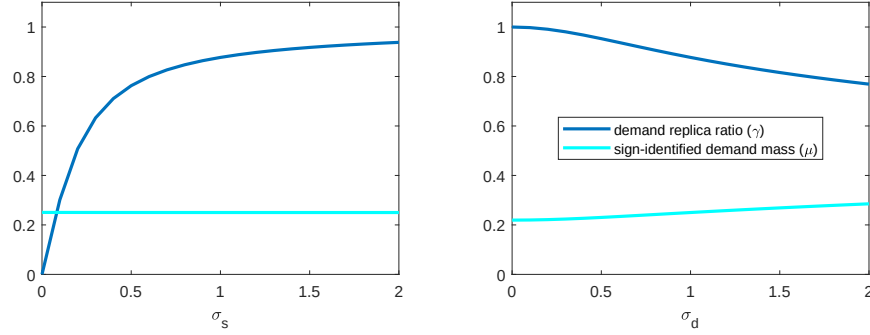
⁸Using the identity $\arccos\left(-\frac{x}{\sqrt{x^2 + y^2}}\right) = \pi - \arctan(y/x)$, we have that $\alpha = \pi - \arctan(\sigma_s/(\omega\sigma_a))$.

the gap widens as σ_s grows. Panel (c) translates this into variance decompositions: the replica attributes a large and increasing share of inflation variance to “demand”—the opposite pattern of the true demand shock. Thus, even in this stylized model, selecting a replica leads the econometrician to overstate the role of demand for inflation.

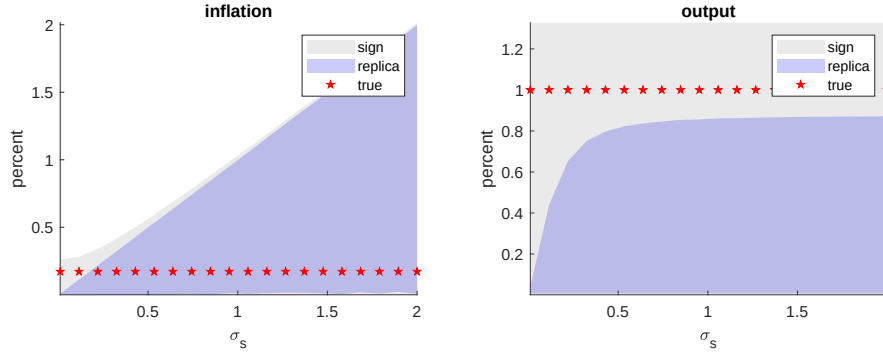
Finally, the sign-identified set is largely uninformative about the role of demand in accounting for the volatility of inflation. When the markup shock standard deviation approaches zero, the contribution of demand can vary from 0 to 70%; with larger values for the markup standard deviation the range widens to 0 to 90%.

These numbers depend on the calibration of parameters and standard deviations. A more quantitative exercise requires a more realistic model and parameter values that match observed features of the data. We turn to this in the next section.

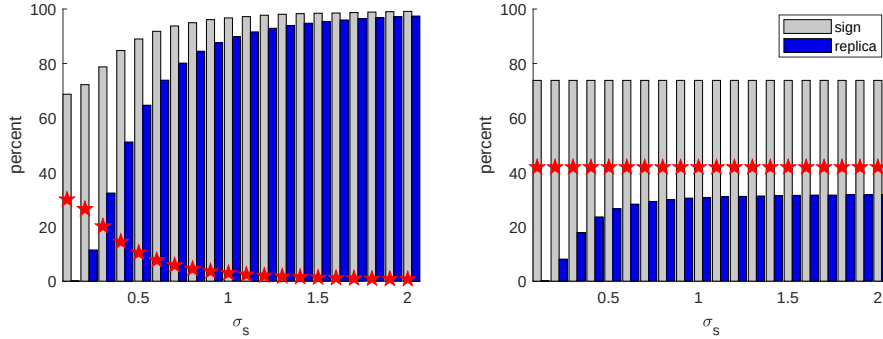
Figure 1: Replicability statistics, impulse responses and variance decomposition of sign-identified demand and replica.



(a) Replicability mass and replica ratio of the demand shock in the three-equation NK model.



(b) Impact of sign-identified demand and replica on inflation and output for different values of σ_s .



(c) Contribution of sign-identified demand and replica to inflation and output variance for different values of σ_s .

Notes: The figure refers to the three-equation NK model of Section 3, identifying the demand shock via Assumption 1. Unless varied on the horizontal axis, parameters are set to $\beta = 0.99$, $\sigma = 1$, $\varphi = 1$, $\theta = 0.75$, $\phi_\pi = 1.5$, $\rho = 0.9$, and $\sigma_a = \sigma_d = 1$. *Panel (a):* admissibility mass μ (light) and replica ratio γ (dark) as functions of the markup volatility σ_s (left, $\sigma_d = 1$) and of the demand volatility σ_d (right, $\sigma_s = 1$). *Panel (b):* impact responses of inflation and output to the sign-identified demand shock and to the demand replica, as a function of σ_s . *Panel (c):* share of the variance of inflation and output attributed to the sign-identified demand shock and to the replica, as a function of σ_s . The replica generates a larger inflation response and a larger inflation variance share, and a smaller output response and output variance share, than the true demand shock.

4 REPLICABILITY OF DEMAND SHOCKS IN NK MODELS

This section studies the replicability of demand shocks in a quantitative New Keynesian (NK) model. We keep the focus on demand shocks, given the interest in their role in explaining business cycle in general and the recent inflation surge in particular. We first assess the extent of replicability with a set of *agnostic* or minimalist restrictions on the sign of the impulse responses of demand and supply shocks. In particular, we compute the demand shock’s *replicability mass* δ , the probability that a zero-weight combination of the remaining shocks reproduces the target sign pattern, and the demand shock’s *replica ratio* γ , which compares that probability with the probability that an unrestricted combination satisfies the same restrictions, and the near-replica density ratio, $\pi_\tau(\varepsilon)$. We then turn to their implication for economic outcomes.

4.1 DATA GENERATING PROCESS

We adopt the medium-scale DSGE model of Smets and Wouters (2007) (henceforth SW) as our theoretical benchmark and data-generating process (DGP). The stochastic dynamics are driven by seven orthogonal disturbances: two demand-side shocks (risk premium and investment-specific technology), three supply-side shocks (total factor productivity, price mark-up, and wage mark-up), a monetary policy shock, and a composite exogenous spending shock (which combines government spending and net export disturbances). Unless differently specified, we assign the estimated posterior mode value to the model parameters as obtained in the original SW paper. Table 1 summarizes the impact signs of output, inflation, and the policy rate across selected structural impulses in the model. We focus on non-policy demand shocks and quantify

Table 1: Impact sign restrictions in SW (selected shocks).

	η^b	η^a	η^p	η^i	$\eta^i + \eta^b$
y	+	+	−	+	+
π	+	−	+	+	+
R	+	−	+	+	+

Notes: The table reports impact signs for output (y), inflation (π), and the policy rate (R) following a demand shock (η^b), a persistent technology shock (η^a), a temporary price mark-up (cost-push) shock (η^p), and an investment-specific technology shock (η^i). The last column reports the combined demand disturbance $\eta^i + \eta^b$.

the extent to which they can be replicated (standalone or in combination) by other

disturbances.⁹

4.2 DEMAND SHOCKS REPLICABILITY

We first restrict attention to the replicability of a single demand shock in a system with three shocks and three observables: two supply shocks and one demand shock driving output, inflation and interest rate. As in the three-equation model discussed in Section 3, the first disturbance is a permanent technology shock, η_t^a , which enters the production function and raises output and depresses inflation. The second is a temporary price mark-up (cost-push) shock, η_t^p , which enters the New Keynesian Phillips curve and raises inflation. The demand shock is the risk-premium disturbance, η_t^b , capturing fluctuations in households' intertemporal preferences. By shifting households' effective discount rate, it acts as a clean IS-type disturbance: it moves demand without directly affecting marginal costs or technology. This provides a neutral benchmark demand shock.

To identify demand shocks in the empirical model, we impose the sign restrictions that have been typically used in the literature to distinguish between demand and supply forces (see, e.g., Giannone and Primiceri (2024)) and are thought to be verified across different models (see, e.g., Canova and Paustian (2011)):

Assumption 2 *A demand shock increases output, inflation, and the policy rate on impact.*

Assumption 3 *A supply shock moves output and inflation in opposite directions on impact.*

The econometrician observes output, inflation, and the policy rate, and seeks to identify demand disturbances with Assumptions 2 and 3. The *ABCD* representation of the model admits a VAR representation, whose impulse responses are known, and coincide with the structural model responses up to an orthogonal rotation. Table 2 reports the replicability mass and the replica ratio, both when the parameters are set to their posterior mode values and when they are drawn uniformly from the prior support (see the appendix). In the latter case, we draw 1,000 parameter vectors uniformly from these supports and recompute the statistics of interest for each draw; the table reports the median along with the minimum and maximum values in brackets.

The replicability mass of the demand shock η^b is 7.2% under Assumption 2, and 2.5% under Assumptions 2–3. The replica ratio is about 43% under Assumption 2

⁹In the appendix we revisit the replicability (*masquerading*) of monetary policy shock which has been discussed in Wolf (2020, 2022) and in Andrade et al. (2025), among others.

Table 2: Replicability mass (δ), replica ratio (γ) and finite- ε replica ratio ($\pi_\tau(\varepsilon)$) of demand shocks in the SW model with three shocks and three observables.

Assumptions	Posterior mode			Uniform Rnd		
	δ	γ	$\pi_\tau(\varepsilon)$	δ	γ	$\pi_\tau(\varepsilon)$
<i>Preference (η^b)</i>						
2	7.2	42.7	43.1	7.2 [7.0, 7.5]	42.4[40.6, 44.1]	42.4[38.5, 47.3]
2+3	2.5	37.4	36.1	2.5 [2.3, 2.6]	37.0 [33.9, 39.5]	37.0[29.8, 44.8]
<i>Preference + investment ($\eta^i + \eta^b$)</i>						
2	7.2	32.8	33.0	7.2 [7.0, 7.5]	32.8[31.4, 34.0]	32.9[28.9, 36.7]
2+3	2.0	26.2	24.5	2.0 [1.8, 2.1]	25.6 [23.7, 27.5]	25.9[21.1, 32.5]

Notes: Values are in percent. “Posterior mode” sets the parameters to the SW posterior mode estimates. “Uniform Rnd” reports the median [min, max] across 1,000 parameter vectors drawn uniformly from the supports reported in the appendix. The top panel refers to the preference shock η^b ; the bottom panel to the composite demand shock $\eta^i + \eta^b$. δ is the replicability mass and $\gamma = \delta/\mu$ the replica ratio, both defined in Section 2. $\pi_\tau(\varepsilon)$ is the finite- ε replica ratio of equation (8), evaluated at $\varepsilon = 0.1$: the frequency of ε -replicas among admissible rotations relative to their frequency under the uniform benchmark, where an ε -replica is a rotation in which the target shock accounts for at most one percent of the variance of the identified disturbance. Since $\pi_\tau(\varepsilon) \rightarrow \gamma$ as $\varepsilon \rightarrow 0$, the proximity of the two columns indicates that the replica ratio is not an artifact of the limiting argument. The corresponding *share* of the identified set made up of ε -replicas is $\pi_\tau(\varepsilon) \Pr(\tau \leq \varepsilon^2)$, which equals $\varepsilon \pi_\tau(\varepsilon)$ for $n = 3$.

alone, and about 37% under Assumptions **2–3**: a *supply replica* of the demand shock (a zero-weight combination of the two supply shocks) satisfies the demand sign pattern roughly 40% as often as an unrestricted combination of the model’s shocks. The last column of each block reports the finite- ε replica ratio $\pi_\tau(\varepsilon)$ of equation (8) at $\varepsilon = 0.1$, and it is essentially indistinguishable from γ : 43.1 against 42.7 under Assumption **2**, and 36.1 against 37.4 under Assumptions **2–3**. The diagnostic therefore does not rely on the limiting argument of Proposition 5: at a tolerance under which the preference shock accounts for at most one percent of the variance of the identified disturbance, the restrictions still favor near-replicas by the same factor. Equivalently, by (9), ε -replicas make up 4.3% of the sign-identified set under Assumption **2** and 3.6% under Assumptions **2–3**, against an unrestricted benchmark of exactly ten percent. The restrictions therefore do reduce the incidence of near-replicas—by roughly a factor of three—without eliminating them: the replica region remains attainable with positive probability, and, as the next figures show, its economic content differs sharply from that of the true demand shock. Thus, even a standard demand–supply sign scheme discriminates only weakly against supply-driven replicas: rotations that contain (almost) no demand shock are penalized far less than proportionally relative to the uniform benchmark (Remark 3). Importantly, these findings are not specific to the posterior-

mode calibration. When we do not take a strong stand on the parameter values and draw them uniformly from wide supports, the median replicability mass and replica ratio are essentially identical to the baseline values, and even the extreme draws move them only marginally (last two columns of Table 2).

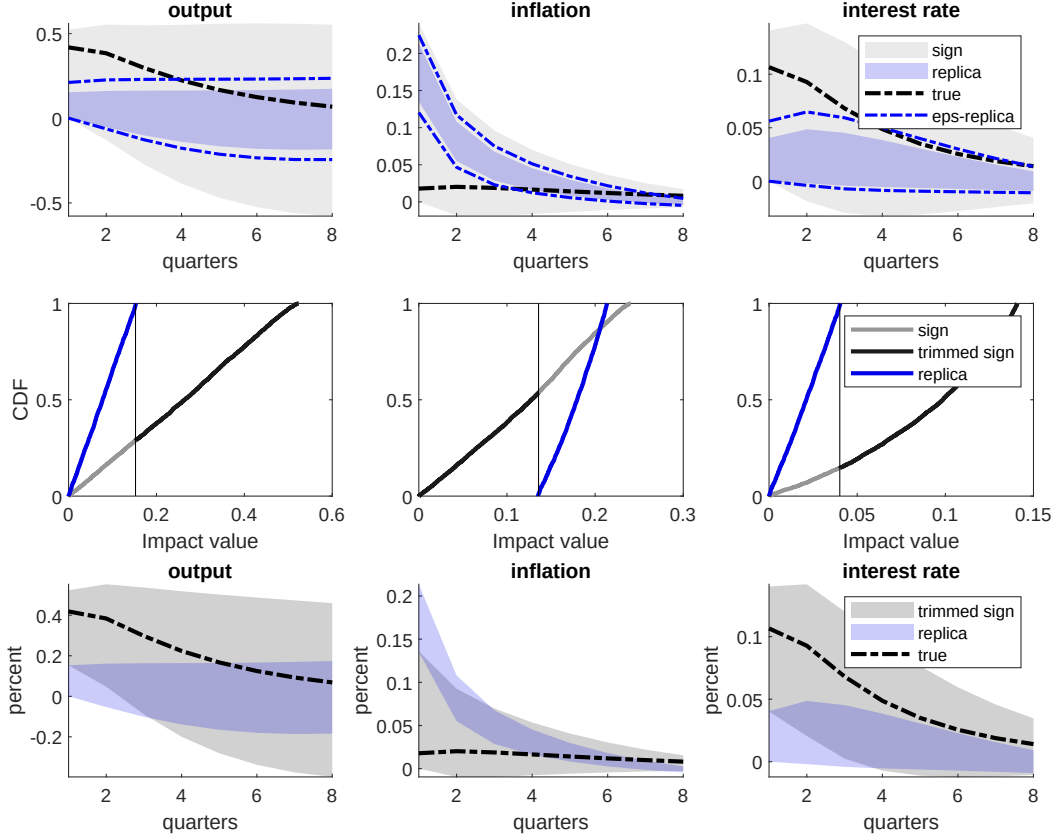
The top row of Figure 2 shows the transmission of the *true* demand shock together with the sign-identified set and the replica set. The replicas feature a substantially larger inflation response, and smaller output and interest-rate responses, than the true demand shock, suggesting that SVARs identified via Assumptions 2–3 may overstate the role of demand in accounting for inflation volatility. Exact replicas place, by construction, zero weight on the preference shock, while the sign-identified set also contains rotations in which this weight is tiny but not zero (the ε -replicas of Remark 3); this explains why the bounds of the replica set need not coincide with those of the sign-identified set. These near-replicas are not a knife-edge. With $\varepsilon = 0.1$ —so that the preference shock accounts for at most one percent of the variance of the identified disturbance—they make up 3.6% of the accepted draws under Assumptions 2–3, against ten percent under the uniform measure with no restrictions imposed, and the finite- ε replica ratio $\pi_\tau(0.1) = 36.1$ is essentially the limiting γ (Table 2). The restrictions thus reduce the incidence of near-replicas without eliminating it, so what follows concerns a region of the identified set that is reached with positive probability rather than an isolated admissible rotation. The middle row of Figure 2 makes the difference between the two sets transparent, plotting the cumulative distribution functions (CDFs) of the impact responses across the sign-identified set and across the replica set: the two distributions overlap only partially, and we exploit this next to refine the identification.

4.3 TRIMMING THE IDENTIFIED SET WITH REPLICA PERCENTILES.

The middle row of Figure 2 suggests a natural refinement: since the impact responses of the replicas are concentrated in a region of the response space that the true demand shock does not visit, one can discard the sign-admissible rotations whose impact responses are consistent with the replica distribution. Formally, let $v_t = \omega' \epsilon_t$ denote a candidate shock, with $\|\omega\| = 1$ and $M\omega > 0$, and let $s_i(\omega)$ denote its implied impact response of variable $i \in \{y, \pi, R\}$. For the replicas $\epsilon_t^r = q' \epsilon_t$, define the cumulative distribution function of the impact response of variable i ,

$$F_i^r(\iota) = \Pr(s_i(q) \leq \iota \mid Mq > 0, \|q\| = 1, e_1' q = 0), \quad (14)$$

Figure 2: Demand shock (η^b): sign-identified set, replicas, ε -replicas, and trimmed set



Notes: The top row depicts the identified set of impulse responses to a demand shock under Assumptions 2–3 (gray), the set of impulse responses of supply replicas satisfying the same restrictions while placing zero weight on the demand shock (blue), the set of ε -replicas, and the responses to the true demand shock (black, dash-dotted). An ε -replica is an admissible rotation in which the preference shock accounts for at most ε^2 , or one percent, of the variance of the identified disturbance; unlike the exact replica set it has strictly positive measure, making up 3.6% of the sign-identified set here against ten percent under the uniform measure with no restrictions imposed (Table 2). The middle row plots the cumulative distribution functions of the impact responses across the sign-identified set (gray) and across the replica set (blue); vertical lines mark the replica percentiles $\iota_{y,1-\alpha}^r$, $\iota_{\pi,\alpha}^r$, and $\iota_{R,1-\alpha}^r$ with $\alpha = 0.01$, defined in equation (14). The bottom row depicts the trimmed identified set of equation (15), obtained by removing the sign-admissible rotations whose impact responses lie within the replica support, together with the replica set and the true responses.

where q is drawn uniformly on the great subsphere $\{q : e_1'q = 0\}$ conditional on the sign restrictions, and let $\iota_{i,\alpha}^r = (F_i^r)^{-1}(\alpha)$ denote its α -percentile. Because replicas are combinations of supply shocks, their output and interest-rate impacts are systematically small, and their inflation impacts systematically large, relative to the sign-identified

set. Given a tail probability α , we therefore retain only the rotations whose impact responses lie outside the support of the replica distribution:

$$\mathcal{G}^{trim}(\alpha) = \{\omega : M\omega > 0, \|\omega\| = 1, s_y(\omega) > \iota_{y,1-\alpha}^r, s_\pi(\omega) < \iota_{\pi,\alpha}^r, s_R(\omega) > \iota_{R,1-\alpha}^r\}. \quad (15)$$

By construction, a replica survives the trimming only if its impact responses fall simultaneously in the three α -tails of the replica distribution, so at most a fraction α of the replica mass survives: for small α , the trimmed set is essentially replica-free.¹⁰

With $\alpha = 0.01$, the replica percentiles $\iota_{y,0.99}^r$, $\iota_{\pi,0.01}^r$, and $\iota_{R,0.99}^r$ correspond to the 28.8th, 53.6th, and 14.5th percentiles of the sign-identified distribution of the impact responses of output, inflation, and the interest rate, respectively (vertical lines in the middle row of Figure 2). Trimming therefore discards the sign-admissible draws whose output impact falls below its 28.8th percentile, whose inflation impact exceeds its 53.6th percentile, or whose interest-rate impact falls below its 14.5th percentile. The bottom row of Figure 2 reports the trimmed identified set: it is tighter than the sign-identified set in the top row and hews more closely to the true transmission, most visibly for inflation, whose upper bound drops substantially. Like the diagnostics of Section 2, the procedure requires a reference model to compute the replica percentiles.

4.4 DISTORTION OF THE DEMAND CONTRIBUTION

The distortions documented so far concern impulse responses; we now quantify their implications for variance decompositions. Figure 3 reports the share of the forecast error variance of output, inflation, and the interest rate attributed to the identified demand shock at the one-quarter (1Q), one-year (1Y), and twenty-five-year (25Y) horizons. For each variable and horizon, the bars report the range of the demand contribution across accepted rotations under the agnostic scheme (Assumptions 2–3, light gray), across replicas (blue), across ε -replicas (with $\varepsilon = 0.1$), and across the trimmed set of equation (15) (dark gray); red stars mark the contribution of the true demand shock in the DGP.

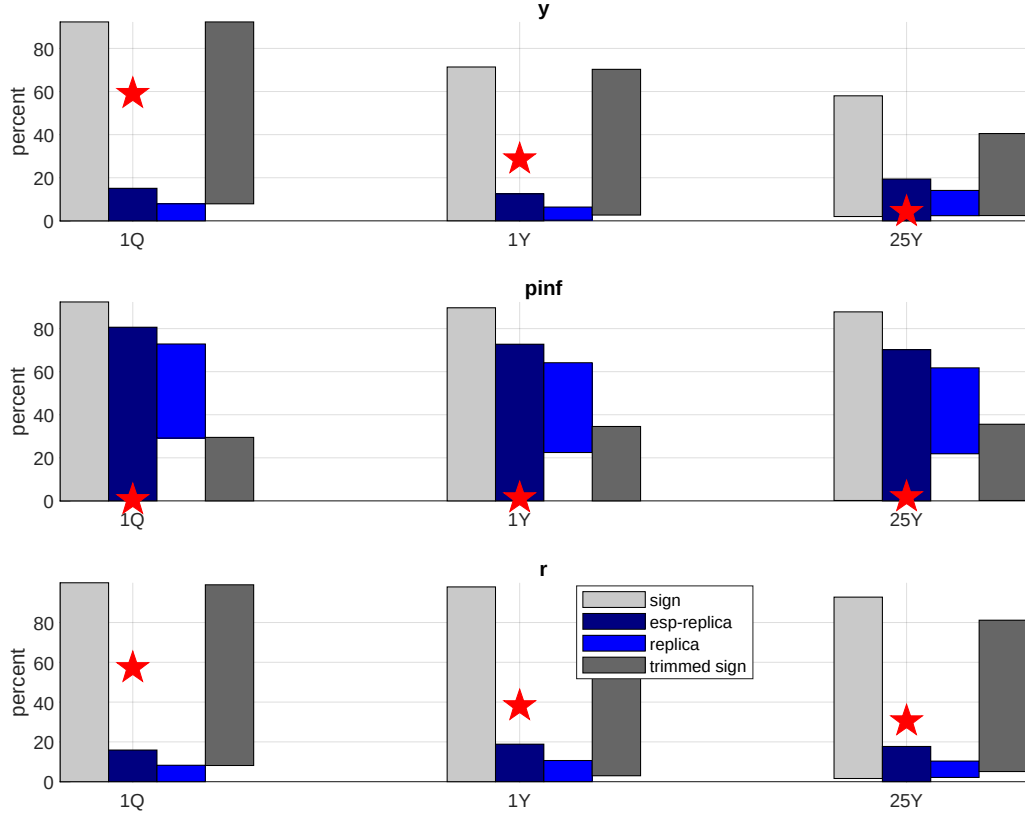
Three results stand out. First, the agnostic sign-identified sets are close to uninfor-

¹⁰Trimming, like any accept–reject implementation of sign restrictions, amounts to conditioning the uniform measure over rotations on a smaller admissible set. The resulting truncated distribution is proper once renormalized by $\Pr(\mathcal{G}^{trim})$, and this renormalization is automatic when statistics (quantiles, means, bands) are computed across the retained draws only. In the empirical application of Section 5, where reduced-form parameters are also drawn, acceptance rates may differ across parameter draws, so pooled summaries implicitly assign more weight to parameterizations that are more compatible with the restrictions—as is the case under standard sign restrictions.

mative: for all variables and horizons, the range of demand contributions spans nearly the entire admissible interval. Second, replicas invert the transmission mechanism. In the model, the true demand shock accounts for a large share of output and interest-rate fluctuations at business-cycle horizons (about 60 percent on impact) and almost none of inflation (1–2 percent). Replicas display the mirror image: they attribute between roughly 30 and 70 percent of inflation variability to “demand” on impact—and only slightly less at longer horizons—while explaining essentially none of output and the interest rate. A researcher selecting a rotation in the replica region would thus conclude that demand drives inflation but not real activity: the opposite of the data-generating process. Nor is that region reached only along the boundary of the identified set. Exact replicas have measure zero, but the ε -replicas of equation (9) do not: at $\varepsilon = 0.1$ they account for 3.6% of the accepted draws under Assumptions 2–3, against ten percent under the uniform measure with no restrictions imposed (Table 2), and they carry essentially the same economic content as the exact replicas whose responses are plotted here.

Two conclusions follow and it is worth keeping them apart. The first is that the replica region is attainable with positive probability, so the pattern documented here is a property of the identified set rather than a curiosity confined to its edge. The second is negative: it does *not* follow that a posterior summary is displaced by a large amount. A subset of mass 3.6% contributes only a few percentage points directly to a posterior mean, and the restrictions in fact reduce the incidence of near-replicas relative to the uniform benchmark. Whether the reported decomposition is materially wrong therefore depends on the distribution of the target loading across the *whole* accepted set, not on its lower tail, and is a question we take up separately. Third, trimming removes most of this distortion. The upper bound of the demand share of inflation falls from about 90 percent to roughly one third, and the lower bounds of the output and interest-rate shares rise above the replica range, while the trimmed sets still contain the true contributions at every horizon. It is important to stress, however, what trimming does *not* deliver: the trimmed bars remain wide. Removing the replica region tightens the upper end of the inflation contribution and the lower ends of the output and interest-rate contributions, but the surviving sign-identified set still spans a broad range at every horizon. In other words, trimming corrects the systematic bias toward demand without turning the sign restrictions into a sharp identification device—the decompositions stay fairly uninformative.

Figure 3: Forecast error variance attributed to the demand shock: sign-identified set, replicas, ε -replicas, and trimmed set



Notes: The figure reports the share of the forecast error variance of output (y), inflation (π), and the interest rate (r) attributed to the identified demand shock at the one-quarter (1Q), one-year (1Y), and twenty-five-year (25Y) horizons. Bars span the range of contributions across accepted rotations: sign-identified set under Assumptions 2–3 (light gray), replica set (blue), ε -replica set, and trimmed set of equation (15) with $\alpha = 0.01$ (dark gray). An ε -replica is an admissible rotation in which the demand shock accounts for at most one percent of the variance of the identified disturbance; unlike the exact replica set it carries strictly positive measure (3.6% of the sign-identified set, against ten percent unrestricted; Table 2), so the contributions it spans are reached with positive probability rather than only along the boundary of the identified set. Red stars mark the contribution of the true demand shock in the data-generating process.

4.5 ROBUSTNESS

Composite Demand shock We next consider the replicability of a composite demand disturbance defined as the sum of the preference and investment-specific shocks, $\eta^i + \eta^b$. The model thus consists of three shocks, one composite demand shock and two supply shocks. More precisely, the core model equations are unchanged; the only

modification are the exogenous processes of the risk premium and investment specific shocks, which are now given by

$$\begin{aligned}\epsilon_t^b &= \rho_b \epsilon_{t-1}^b + \sigma_b \eta_t, \\ \epsilon_t^i &= \rho_i \epsilon_{t-1}^i + \sigma_i \eta_t, \\ \eta_t &\sim N(0, 1)\end{aligned}$$

where the same innovation (η_t) affects the two exogenous processes and the two wedges in the respective equilibrium conditions. As before, the econometrician observes output, inflation, and the policy rate. The modified *ABCD* representation of the model still admits a VAR representation, whose impulse responses coincide with the structural model responses up to an orthogonal rotation. Assumptions for identification are identical.

Looking at the last two rows of Table 2, the replicability mass of $\eta^i + \eta^b$ is about 7.2% under Assumption 2 and 2.0% under Assumptions 2–3. The replica ratio of the combined demand shock is about 33% under Assumption 2 and about 26% under Assumptions 2–3: combinations constructed exclusively from non-demand disturbances satisfy the demand restrictions roughly a third as often as unrestricted combinations (Table 2, bottom panel). As in the single demand case, the limiting diagnostic survives at a finite tolerance: $\pi_\tau(0.1)$ equals 33.0 and 24.5 against a limiting γ of 32.8 and 26.2, so that ε -replicas occupy 3.3% and 2.5% of the sign-identified set under the two schemes. Pooling the two demand shocks therefore lowers the scope for replication relative to the preference shock alone, but does not eliminate it. As in the single demand case, the replica shock generates large fluctuations in inflation and small fluctuations in output and the interest rate, which are inconsistent with the transmission of the ‘true’ composite demand shock. The values of the trimmed set are: with $\alpha = 0.01$, the replica percentiles $\iota_{y,0.99}^r$, $\iota_{\pi,0.01}^r$, and $\iota_{R,0.99}^r$ correspond to the 14.8th, 46.4th, and 8.2nd percentiles of the sign-identified distribution of the impact responses of output, inflation, and the interest rate, respectively. We discard then rotation that generate an impact for output smaller than the 14.8th percentile, for inflation larger than the 46.4th percentile and for interest rate smaller than the 8.2nd percentile, respectively. While numbers are quantitatively different, the replica IRFs, the trimmed sign IRF and variance decompositions are qualitatively similar to those obtained for the single demand shock, see Figures 6 and 7 in the appendix.

Seven (six) observables and seven (six) shocks. We consider the replicability of the preference (η^b) and composite ($\eta^i + \eta^b$) demand shock defined as above with the original seven observables used in the SW paper. The remaining shocks are price and wage markup, technology, exogenous spending, monetary policy. We assume the same restrictions as in Assumptions 2 and 3 to identify SVAR shocks, and that the econometrician observes output, inflation, interest rate, hours worked, real wages, investment and consumption. As discussed in Section 2.2, replicability becomes more likely when the number of shocks increases relative to the number of restrictions. Using the estimated mode value of the parameters, the replicability mass of η^b is 12% under Assumption 2 and 3% under Assumptions 2–3. The replica ratio of the preference shock is about 83% under Assumption 2 and about 77% under Assumptions 2–3: with seven shocks (and seven observables), combinations that exclude the preference shock satisfy the demand restrictions almost as easily as unrestricted ones. It is virtually impossible to detect a support region where only the replica shock lies as it overlaps almost identically with the preference shock except for consumption, see Figure 8 in the appendix, where the support of the replica shock is mostly negative on impact.

We now turn to the replicability properties of $\eta^i + \eta^b$. As in the previous section we assume that same innovation (η_t) affects the risk premium and investment specific shocks. Moreover, we assume that the econometrician observes output, inflation, interest rate, hours worked, real wages and consumption.¹¹ The replicability mass of $\eta^i + \eta^b$ is 10% under Assumption 2 and 3% under Assumptions 2–3. The replica ratio of the combined demand shock is about 59% under Assumption 2 and 54% under Assumptions 2–3. While these numbers are slightly smaller than in the previous case, the main considerations are similar. The replica shock support overlaps almost identically with the composite demand shock except for consumption and output, see Figure 10 in the appendix, where the support of the replica shock is mostly negative on impact on consumption and smaller for output. Trimming the sign-admissible combinations with this information does not help to improve inference and to correctly identify the contribution of demand to output and inflation volatility.

In all, in the case with six or seven shocks (and an equivalent number of observables) Assumptions 2–3 are insufficient to ensure that the SVAR shocks capture the intended structural disturbances. Moreover, the sign-identified sets are close to uninformative for variance decomposition exercises: for all variables and horizons, the range of demand contributions spans nearly the entire admissible interval making any statement about the role of demand in explaining fluctuations highly unreliable.

¹¹Results are similar using investment instead of consumption.

5 REVISITING THE ROLE OF DEMAND FOR INFLATION DYNAMICS

In this section, we aim to identify the contributions of demand and supply disturbances to the unforecastable variability in GDP and inflation from 2020 onward in the U.S and in the Euro Area, paying particular attention to the 2021–2024 inflation episode, which corresponds to the period following the pandemic crisis and the Ukraine war.

Throughout our empirical analysis, we follow Giannone and Primiceri (2024): we estimate a trivariate (GDP, inflation and interest rate) VAR with 4 lags under agnostic sign restrictions of demand and supply (i.e. Assumptions 2-3) using quarterly data from 1997Q1 to 2019Q4 as estimation sample, and then use the estimated coefficients to decompose unforecasted variability from 2020Q1 to 2024Q1 in demand and supply contributions.

We wish to show the sensitivity of the contribution of demand and supply to these episodes if the econometrician augments standard sign restrictions with trimmed-ones that address misidentification issues arising from the replicability of demand disturbances as discussed in the previous sections. While important and sizable, we ignore the fact that identified set implies almost uninformative bands around the implied contribution and focus on mean decompositions as it is typically done in the literature. Bergholt et al. (2024) is a notable exception. They show that uncertainty is a relevant issue for historical decompositions: coefficient draws associated with IRFs that are very close to the median response can nevertheless generate substantially different historical decompositions. This result is driven by uncertainty surrounding the estimation of the deterministic component of the model. Our point is different: the source of sensitivity is not the uncertainty around parameters estimates rather the un informativeness derived from the identifying assumptions.

To address their concerns, we use the sum-of-coefficients prior. Moreover, we compute one historical decomposition for each of our 5,000 parameter draws and report the pointwise mean historical decomposition at each point in time. This solution is similar to the proposal in Bergholt et al. (2024), who compute a large number of historical decompositions, each associated with a parameter draw, and recover the pointwise median historical decomposition at each point in time.¹² We use the mean rather than the median in order to preserve the additivity property.

¹²In the words of the authors, 'While such a summary measure mixes information from different draws and does not preserve the additivity property of historical decompositions, it features the non-negligible advantage of taking into account the uncertainty in the deterministic component into the inferential process while insuring robustness against non-normalities in the distribution of estimates.'

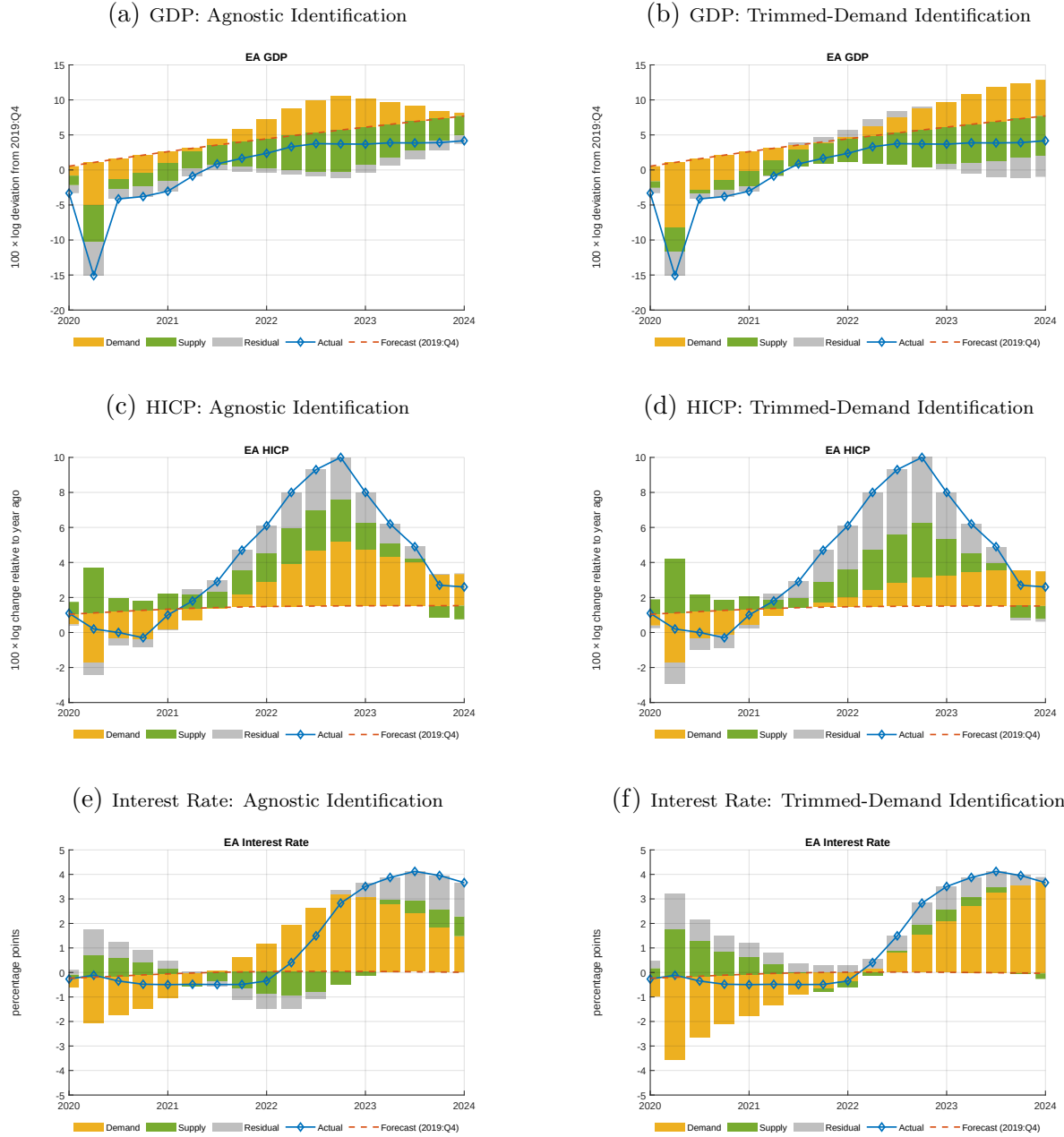
5.1 CONTRIBUTION OF DEMAND AFTER TRIMMING

As in Giannone and Primiceri (2024) we estimate a trivariate SVAR with (log) real GDP and inflation (CPI for the U.S., HICP for the Euro Area), and the interest rates, i.e. the 1-year Euribor for the euro area and the 1-year Treasury yield for the U.S. Under the agnostic scheme, demand disturbances are restricted to move GDP, inflation, and the interest rate in the same direction, whereas supply disturbances are restricted to induce a negative comovement between output and inflation.

In the refined demand identification, we rule out from the set of demand disturbances rotations that are indistinguishable from supply replica; i.e. we complement the agnostic restriction by trimming the responses using the replica percentiles as discussed in Section 4.3 where demand shocks are a combination of preference or risk-premium shocks and investment specific shocks. In particular, for each draw from the posterior distribution of the parameters, we generate 1,000 admissible rotations, and trim them discarding the sign-admissible draws whose output impact after a demand shock falls below its 14.8th percentile, whose inflation impact exceeds its 46.4th percentile, or whose interest-rate impact falls below its 8.2nd percentile, and retain randomly one admissible rotation per reduced form parameter draw.

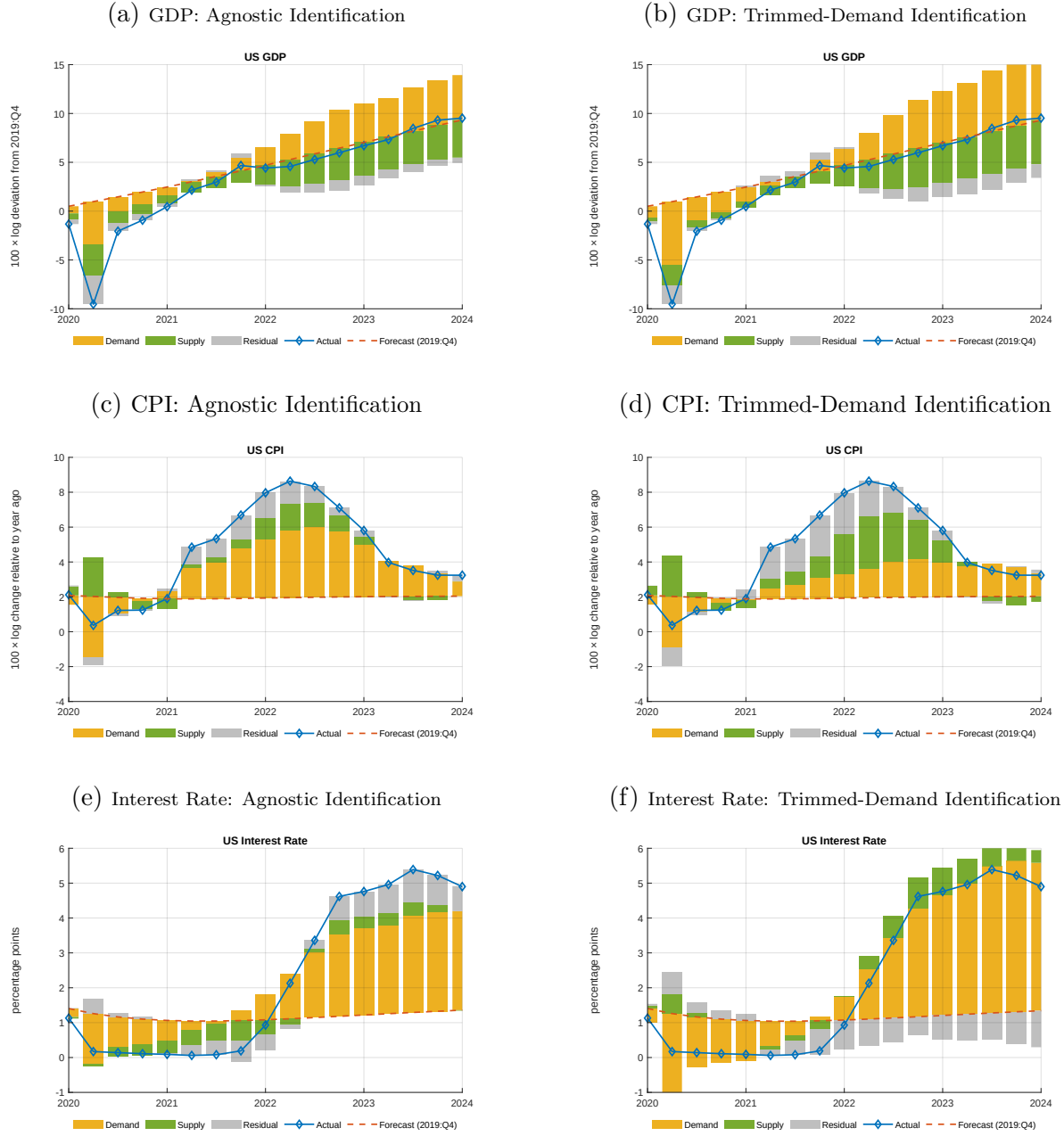
Figures 4 and 5 report the historical decompositions obtained under the agnostic and demand-trimmed identification schemes for the euro area and the U.S., respectively. Trimming substantially changes the attribution of the 2021–2024 inflation episode. Once demand shocks are refined, the contribution of supply disturbances increases and that of demand declines in both economies. By contrast, the historical decompositions of GDP remain broadly similar across the two identification schemes, albeit the effect of demand appear more persistent. The decomposition of interest rates changes more visibly: unforecasted interest rate variation becomes more demand-driven.

Figure 4: Supply vs. demand contributions to GDP, HICP, and interest rate dynamics in the Euro Area.



Notes: The figure reports forecast error decompositions under agnostic and supply-refined identification schemes in a three-variable system. Bars represent the contribution of structural shocks to deviations from the no-shock forecast. Comparing columns highlights how demand refinement alters the attribution of macroeconomic fluctuations.

Figure 5: Supply vs. demand contributions to GDP, CPI, and interest rate dynamics in the United States.



Notes: The figure reports forecast error decompositions under agnostic and supply-refined identification schemes in a three-variable system. Bars represent the contribution of structural shocks to deviations from the no-shock forecast. Comparing columns highlights how demand refinement modifies the attribution of macroeconomic fluctuations.

6 CONCLUSIONS

This paper studies replicability in sign-restricted SVARs: the possibility that structural disturbances other than the target shock generate the same identifying sign pattern. We formalize this problem through the notions of replica shocks, replicability mass, replica ratio and ε -replica share. These objects provide a quantitative measure of how informative a sign-restriction scheme really is. When the admissible set contains many no-self-use combinations that satisfy the target signs, sign restrictions alone cannot guarantee that the identified disturbance corresponds to the intended economic shock.

Our model-based results show that this issue is economically important. In a standard New Keynesian environment, agnostic sign restrictions admit non-policy replicas of monetary policy shocks and—novel to this paper—supply-driven replicas of demand shocks. These replicas can imply markedly different transmission mechanisms and materially distort forecast error variance decompositions. Crucially, the misleading region is not confined to the measure-zero set of exact replicas: near-replicas carrying the same economic content are reached with positive probability, so replicability is a property of the identified set rather than of its boundary alone. In particular, standard demand-supply sign schemes may overstate the role of demand in explaining inflation fluctuations.

We also show that the problem can be mitigated. Restrictions on the relative magnitude of inflation and output responses using percentiles of the replica distribution sharply reduce the scope for demand replicas. In the empirical application to post-2020 inflation in the United States and the Euro Area, refined identification shifts part of the inflation contribution away from demand and partly toward supply. This mitigation has a caveat, however. Even after refinement, the decompositions implied by these sign restrictions remain wide: trimming removes the replica-driven bias but does not make the sign restrictions sharply informative about the demand contribution.

Sign restrictions are often presented as weak assumptions, but, given a structural model as a reference, weak assumptions need not be weakly consequential. If they are too permissive, they may admit economically misleading structural interpretations. For this reason, applied work should treat the informativeness of sign restrictions as an object to be measured rather than presumed. Reporting replicability diagnostics, and designing restrictions with replicas in mind, can improve transparency and make structural conclusions more credible.

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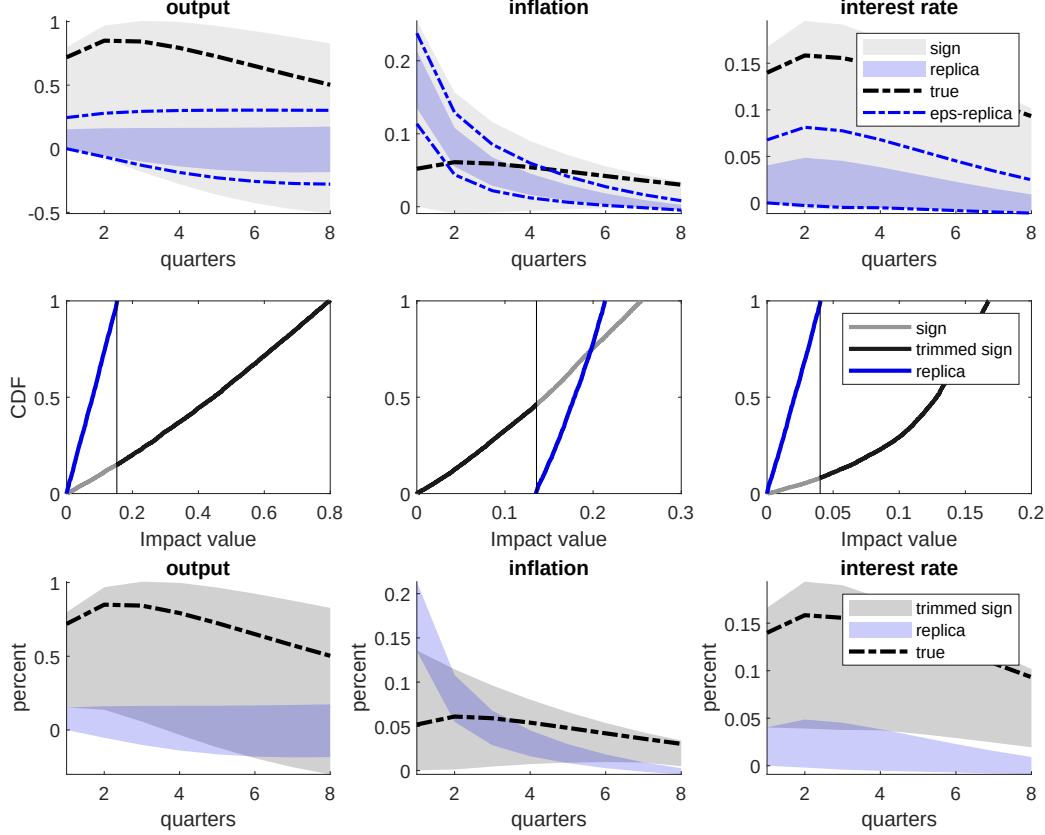
A ADDITIONAL GRAPHS AND TABLES

Table 3: Supports of the uniform draws over the Smets–Wouters parameters

Parameter	Description	Support
<i>Preferences and technology</i>		
α	Capital share	[0.15, 0.40]
ψ	Capacity utilization cost	[0.05, 0.70]
σ_c	Risk aversion	[0.50, 2.00]
σ_l	Frisch elasticity	[0.50, 2.00]
λ	External habit degree	[0.01, 0.90]
<i>Nominal rigidities</i>		
ξ_p	Calvo parameter, prices	[0.01, 0.90]
ξ_w	Calvo parameter, wages	[0.01, 0.90]
ι_p	Indexation to past prices	[0.01, 0.80]
ι_w	Indexation to past wages	[0.01, 0.80]
<i>Monetary policy rule</i>		
r_π	Inflation feedback	[1.05, 2.50]
r_y	Output level feedback	[0.01, 0.50]
ρ	Interest rate persistence	[0.05, 0.90]
<i>Shock persistence</i>		
ρ_a	Total factor productivity	[0.50, 0.95]
ρ_b	Risk premium	[0.05, 0.75]
ρ_i	Investment-specific	[0.05, 0.75]
ρ_p	Price mark-up	[0.05, 0.75]
μ_p	Price mark-up, MA term	[0.05, 0.75]
<i>Shock standard deviations</i>		
σ_a	Total factor productivity	[0.05, 0.70]
σ_b	Risk premium	[0.05, 0.70]
σ_p	Price mark-up	[0.05, 0.70]
σ_i	Investment-specific	[0.05, 0.70]

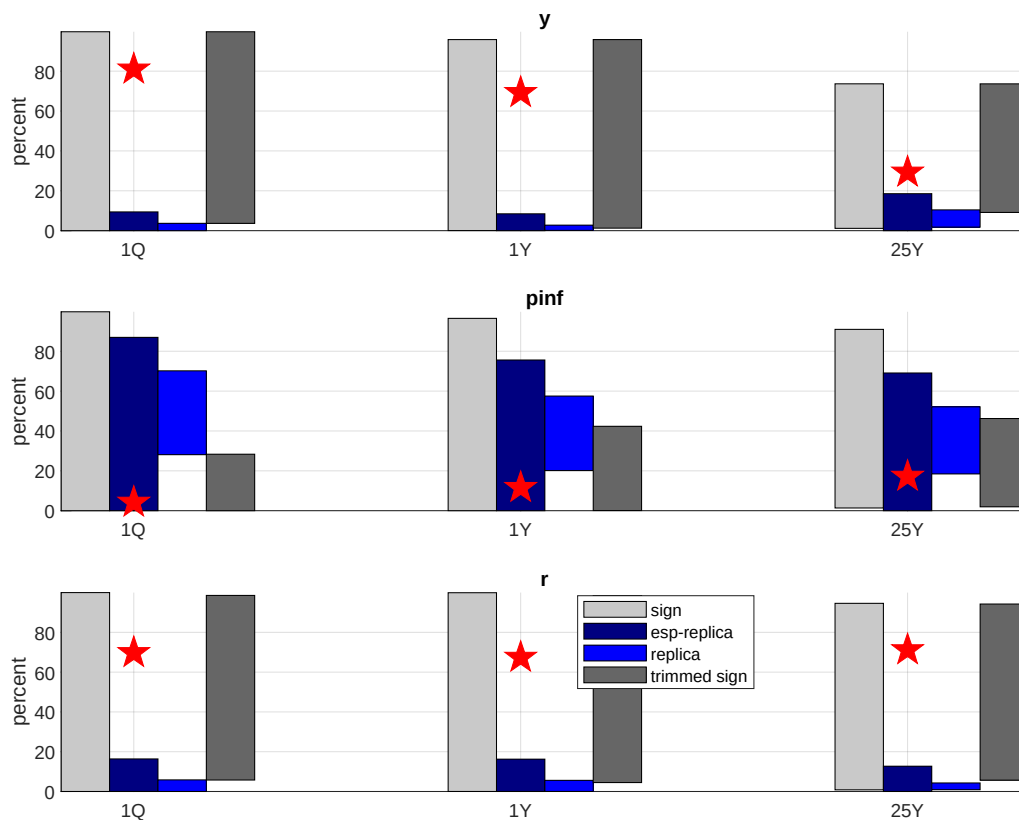
Notes: The table reports the supports of the independent uniform distributions from which the 1,000 parameter vectors underlying the “Uniform Rnd” columns of Table 2 are drawn. Notation and parameter names follow Smets and Wouters (2007). All parameters not listed are held at the Smets and Wouters (2007) posterior mode; these include the investment adjustment cost, the fixed-cost share, the Kimball aggregator curvatures, the depreciation rate, the steady-state growth, inflation, hours and discount rate, the persistence of the exogenous spending, monetary policy and wage mark-up shocks together with the MA coefficient of the latter, and the standard deviations of the exogenous spending, monetary policy and wage mark-up shocks. The lower bound on r_π imposes the Taylor principle. Within this table λ denotes the external habit degree and ρ the smoothing coefficient of the policy rule, following Smets and Wouters (2007); elsewhere in the paper the same symbols denote the slope of the Phillips curve and the persistence of the technology process of Section 3.

Figure 6: Composite demand shock ($\eta^b + \eta_t^i$): sign-identified set, replicas, ε -replicas, and trimmed set



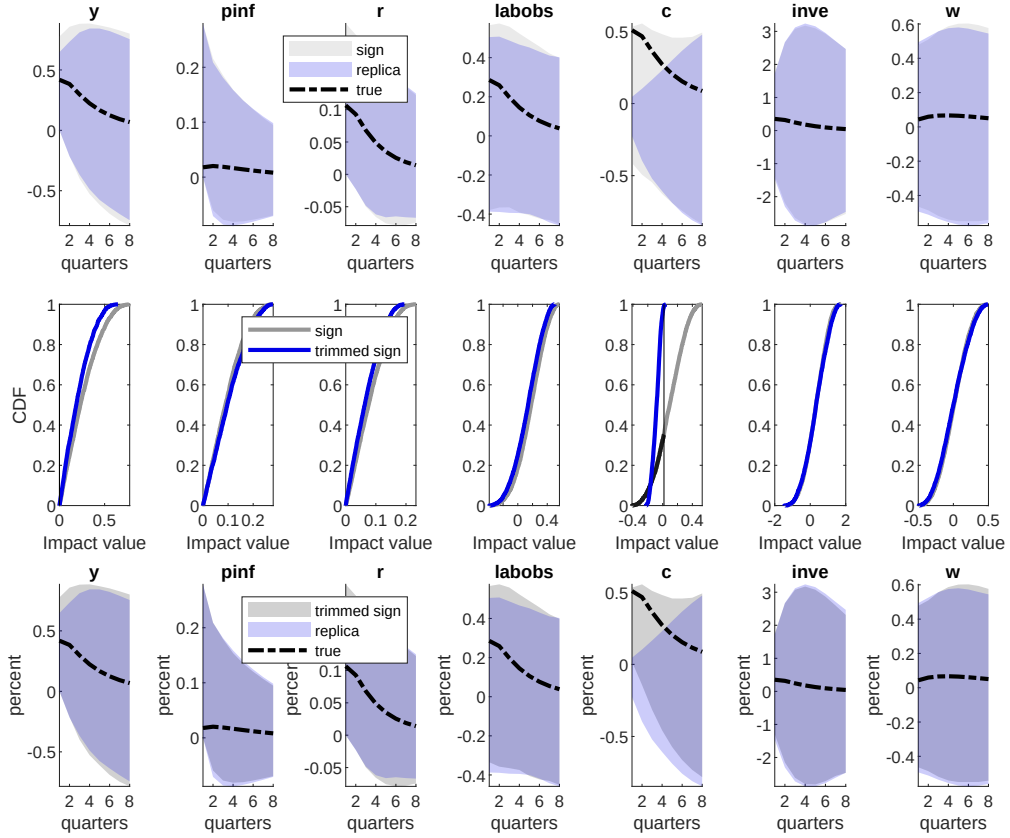
Notes: The top row depicts the identified set of impulse responses to a composite demand shock under Assumptions 2–3 (gray), the set of impulse responses of supply replicas satisfying the same restrictions while placing zero weight on the demand shock (blue), the set of ε -replicas, and the responses to the true demand shock (black, dash-dotted). An ε -replica is an admissible rotation in which the composite demand shock accounts for at most ε^2 , or one percent, of the variance of the identified disturbance; unlike the exact replica set it has strictly positive measure, making up 2.5% of the sign-identified set here against ten percent under the uniform measure with no restrictions imposed (Table 2). The middle row plots the cumulative distribution functions of the impact responses across the sign-identified set (gray) and across the replica set (blue); vertical lines mark the replica percentiles $\iota_{y,1-\alpha}^r$, $\iota_{\pi,\alpha}^r$, and $\iota_{R,1-\alpha}^r$ with $\alpha = 0.01$, defined in equation (14). The bottom row depicts the trimmed identified set of equation (15), obtained by removing the sign-admissible rotations whose impact responses lie within the replica support, together with the replica set and the true responses.

Figure 7: Forecast error variance attributed to the composite demand shock: sign-identified set, replicas, ε -replicas, and trimmed set



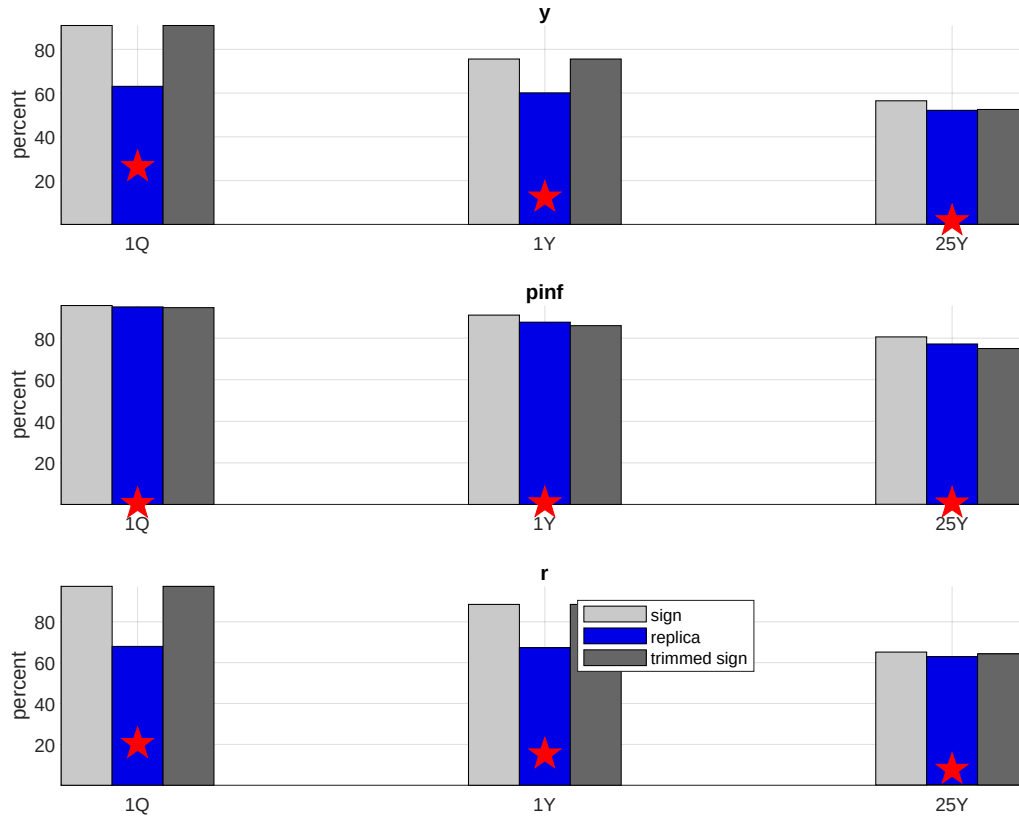
Notes: The figure reports the share of the forecast error variance of output (y), inflation (π), and the interest rate (r) attributed to the identified composite demand shock at the one-quarter (1Q), one-year (1Y), and twenty-five-year (25Y) horizons. Bars span the range of contributions across accepted rotations: sign-identified set under Assumptions 2–3 (light gray), replica set (blue), ε -replica set, and trimmed set of equation (15) with $\alpha = 0.01$ (dark gray). An ε -replica is an admissible rotation in which the composite demand shock accounts for at most one percent of the variance of the identified disturbance; unlike the exact replica set it carries strictly positive measure (2.5% of the sign-identified set, against ten percent unrestricted; Table 2), so the contributions it spans are reached with positive probability rather than only along the boundary of the identified set. Red stars mark the contribution of the true demand shock in the data-generating process.

Figure 8: Demand shock (η^b) in the 7 shocks 7 observables model: sign-identified set, replicas, and trimmed set



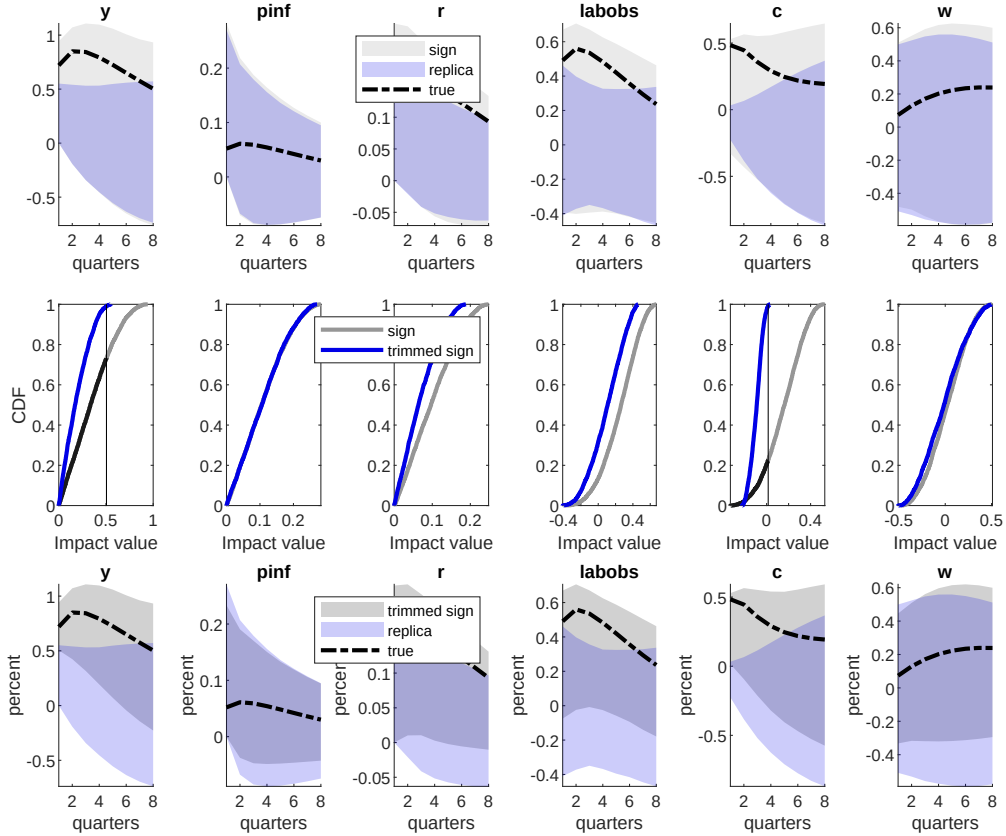
Notes: The top row depicts the identified set of impulse responses to the demand shock under Assumptions 2–3 (gray), the set of impulse responses of supply replicas satisfying the same restrictions while placing zero weight on the demand shock (blue), and the responses to the true demand shock (black, dash-dotted). The middle row plots the cumulative distribution functions of the impact responses across the sign-identified set (gray) and across the replica set (blue); vertical lines mark the replica percentiles $\iota_{c,1-\alpha}^r$ with $\alpha = 0.01$, defined in equation (14). The bottom row depicts the trimmed identified set of equation (15), obtained by removing the sign-admissible rotations whose impact responses lie within the replica support, together with the replica set and the true responses.

Figure 9: Forecast error variance attributed to the demand shock (η^b) in the 7 shocks 7 observables model: sign-identified set, replicas, and trimmed set



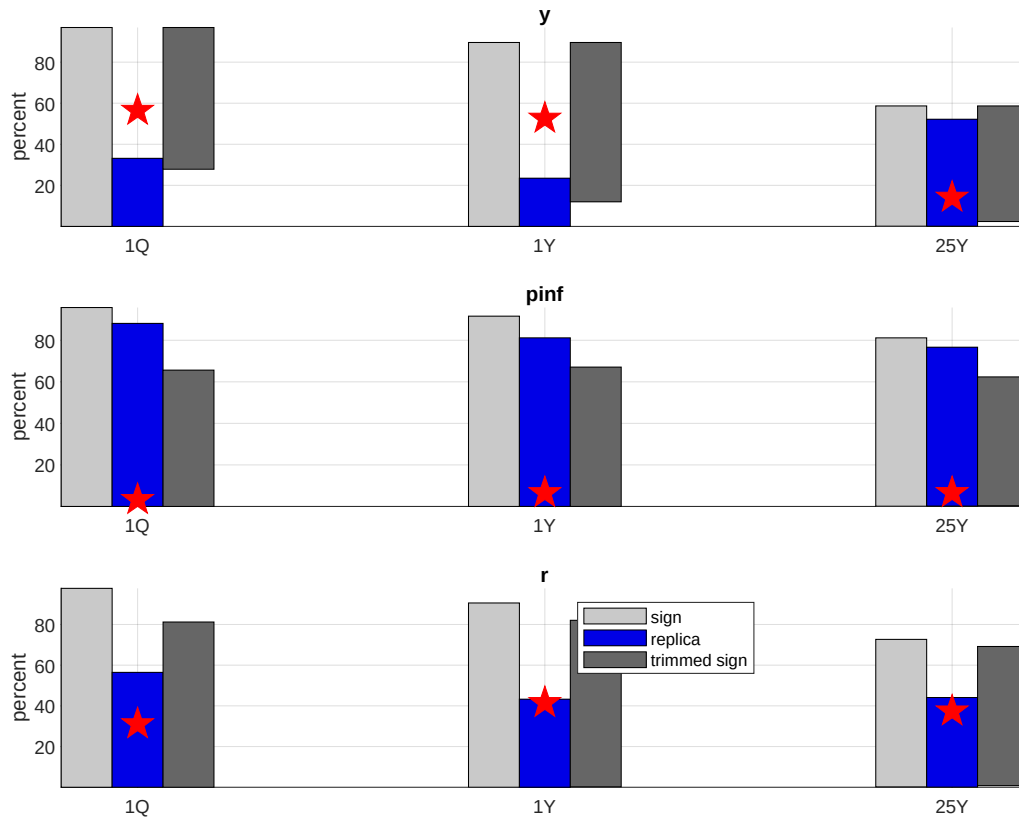
Notes: The figure reports the share of the forecast error variance of output (y), inflation (π), and the interest rate (r) attributed to the identified demand shock at the one-quarter (1Q), one-year (1Y), and twenty-five-year (25Y) horizons. Bars span the range of contributions across accepted rotations: sign-identified set under Assumptions 2–3 (light gray), replica set (blue), and trimmed set of equation (15) with $\alpha = 0.01$ (dark gray). Red stars mark the contribution of the true demand shock in the data-generating process.

Figure 10: Composite demand shock ($\eta^b + \eta^i$) in the 6 shocks 6 observables model: sign-identified set, replicas, and trimmed set



Notes: The top row depicts the identified set of impulse responses to the demand shock under Assumptions 2–3 (gray), the set of impulse responses of supply replicas satisfying the same restrictions while placing zero weight on the demand shock (blue), and the responses to the true demand shock (black, dash-dotted). The middle row plots the cumulative distribution functions of the impact responses across the sign-identified set (gray) and across the replica set (blue); vertical lines mark the replica percentiles $\iota_{y,1-\alpha}^r$ and $\iota_{c,1-\alpha}^r$ with $\alpha = 0.01$, defined analogously to equation (14). The bottom row depicts the trimmed identified set of equation (15), obtained by removing the sign-admissible rotations whose impact responses lie within the replica support, together with the replica set and the true responses.

Figure 11: Forecast error variance attributed to the composite demand shock in the 6 shocks 6 observables model: sign-identified set, replicas, and trimmed set



Notes: The figure reports the share of the forecast error variance of output (y), inflation (π), and the interest rate (r) attributed to the identified demand shock at the one-quarter (1Q), one-year (1Y), and twenty-five-year (25Y) horizons. Bars span the range of contributions across accepted rotations: sign-identified set under Assumptions 2–3 (light gray), replica set (blue), and trimmed set of equation (15) with $\alpha = 0.01$ (dark gray). Red stars mark the contribution of the true demand shock in the data-generating process.

B PROOF OF PROPOSITIONS

Proposition 5: Proof. The uniform measure on $\mathbb{S}^{n-1}$ disintegrates over the hyperplane slices $\{q_1 = x\}$: writing $q = (x, \sqrt{1-x^2}u)$ with $u \in \mathbb{S}^{n-2}$, the surface measure factorizes as $(1-x^2)^{(n-3)/2}dx d\sigma(u)$, so the marginal density of q_1 is $f_n(x) = c_n(1-x^2)^{(n-3)/2}$, with $f_n(0) = c_n$ as defined in (6). Conditional on $q_1 = x$ the remaining coordinates are uniform on a sphere of radius $\sqrt{1-x^2}$, so $\Pr(Mq > 0 \mid q_1 = x) \rightarrow \delta$ as $x \rightarrow 0$ under the stated transversality condition. Hence

$$\Pr(\tau(q) \leq \varepsilon^2, q \in \mathcal{G}) = \int_{-\varepsilon}^{\varepsilon} \Pr(Mq > 0 \mid q_1 = x) f_n(x) dx = \delta \cdot 2c_n \varepsilon \cdot (1 + o(1)).$$

Dividing by $\mu = \Pr(q \in \mathcal{G})$ and by $\Pr(\tau(q) \leq \varepsilon^2) = 2c_n \varepsilon (1 + o(1))$ gives the result. The argument is carried out in q_1 rather than in τ because $\tau = q_1^2$ identifies the slices at $+x$ and $-x$, on which the admissible cone generally has different mass; conditioning on τ would therefore not deliver δ in the limit. ■

Proposition 6: Proof. (i) $\mathcal{F}$ is the intersection of the open cone $\mathcal{K}$ with the great subsphere, hence relatively open in $\mathbb{S}^{n-2}$; a nonempty relatively open subset of $\mathbb{S}^{n-2}$ has strictly positive surface measure, so $\delta = 0$ forces $\mathcal{F} = \emptyset$. The converse is immediate, as is the equivalence with $\gamma = 0$ given $\mu > 0$. (ii) If $\delta > 0$ then $\mathcal{F} \neq \emptyset$ by (i), so some $q \in \mathcal{G}$ has $e'_1 q = 0$ and hence $\tau(q) = 0$; since $\tau \geq 0$ the infimum equals zero and is attained at that q . (iii) The first claim is the contrapositive of (ii). For the second, $\mathcal{G}$ is open, so its closure may meet $\mathcal{H}$ without $\mathcal{G}$ doing so; along any sequence in $\mathcal{G}$ approaching such a tangency point, $\tau \rightarrow 0$ while $\mathcal{F}$ remains empty. ■

Proposition 7: Proof. *Step 1: the impact matrix.* First combine the MP rule with the DIS, and get

$$y_t = E_t y_{t+1} - \frac{\phi_\pi - 1}{\sigma} E_t \pi_{t+1} + \sigma_d \epsilon_t^d.$$

Guess the solution, i.e. $\pi_t = c_\pi a_{t-1}$, $y_t = c_y a_{t-1}$, $i_t = c_i a_{t-1}$. Assume zero shocks, i.e. $\sigma_a = \sigma_s = \sigma_d = 0$. Then we have, $E_t \pi_{t+1} = c_\pi \rho a_{t-1}$, $E_t y_{t+1} = c_y \rho a_{t-1}$, $E_t i_{t+1} = c_i \rho a_{t-1}$. Moreover, the non-policy block becomes:

$$\begin{aligned} c_\pi &= \beta c_\pi \rho + \lambda(\sigma + \varphi) c_y - \lambda(1 + \varphi) \rho, \\ c_y &= \rho c_y - \frac{\phi_\pi - 1}{\sigma} \rho c_\pi. \end{aligned}$$

which are verified when

$$c_\pi = -\rho\omega \quad c_y = -\rho \frac{\phi_\pi - 1}{\sigma(1 - \rho)} c_\pi \quad c_i = \phi_\pi \rho c_\pi,$$

where $\chi = 1 - \beta\rho + \lambda(\sigma + \varphi)(\phi_\pi - 1)\rho/(\sigma(1 - \rho)) > 0$, $\omega = \frac{\lambda(1+\varphi)}{\chi} > 0$. To derive the impact matrix, we follow the approach of Uhlig (1999). Let $\epsilon_t = \begin{pmatrix} \epsilon_t^d & \epsilon_t^a & \epsilon_t^s \end{pmatrix}'$ and $x_t = \begin{pmatrix} \pi_t & y_t & a_t \end{pmatrix}'$. The NK model can be cast in

$$0 = E_t(Fx_{t+1} + Gx_t + Hx_{t-1} + L\epsilon_t),$$

where

$$F = \begin{pmatrix} \beta & 0 & 0 \\ -\frac{\phi_\pi - 1}{\sigma} & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, G = \begin{pmatrix} -1 & \lambda(\varphi + \sigma) & -\lambda(1 + \varphi) \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, H = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \rho \end{pmatrix}, L = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

and the solution takes the form of $x_t = Px_{t-1} + Q\epsilon_t$. The matrix P solves $FP^2 + GP + H = 0$, with

$$P = \begin{pmatrix} 0 & 0 & c_\pi \\ 0 & 0 & c_y \\ 0 & 0 & \rho \end{pmatrix}.$$

Then, the impact matrix is given by

$$Q = -(FP + G)^{-1}L = \begin{pmatrix} \lambda(\varphi + \sigma) & -\omega & 1 \\ 1 & \eta & 0 \\ 0 & 1 & 0 \end{pmatrix},$$

where $\eta = \frac{(\phi_\pi - 1)\rho}{\sigma(1 - \rho)}\omega > 0$. The model $ABCD$ representation in terms of π_t , y_t and i_t as observables follows immediately. Since the structural shocks enter the model scaled by their standard deviations, the impact responses of inflation and output to ϵ_t are the first two rows of $Q \text{diag}(\sigma_d, \sigma_a, \sigma_s)$, that is m_1 and m_2 as defined in the statement. A candidate SVAR demand shock $\nu_t^d = q'\epsilon_t$ therefore satisfies Assumption 1 if and only if $m_1q > 0$ and $m_2q > 0$.

Step 2: the replicability mass. Imposing $q_1 = 0$, the two inequalities reduce to $-\omega\sigma_a q_2 + \sigma_s q_3 > 0$ and $\eta\sigma_a q_2 > 0$, i.e. to $q_2 > 0$ and $q_3 > \frac{\omega\sigma_a}{\sigma_s} q_2$. In polar coordinates, $q_2 = \cos \theta$ and $q_3 = \sin \theta$, these read $\tan \theta > \frac{\omega\sigma_a}{\sigma_s}$ and $\cos \theta > 0$, so that

$$\theta \in \left(\arctan \left(\frac{\omega\sigma_a}{\sigma_s} \right), \frac{\pi}{2} \right).$$

The length of this arc is $\pi/2 - \arctan(\omega\sigma_a/\sigma_s) = \arctan(\sigma_s/(\omega\sigma_a))$, using $\pi/2 - \arctan x = \arctan(1/x)$ for $x > 0$. Dividing by the circumference 2π of the great circle $\{q \in \mathbb{S}^2 : q_1 = 0\}$ delivers δ .

Step 3: the admissibility mass. Without the zero-weight constraint, $\mathcal{G} = \{q \in \mathbb{S}^2 : m_1 q > 0, m_2 q > 0\}$ is the intersection of two hemispheres, i.e. a lune whose dihedral angle is $\pi - \alpha$, where α is the angle between the normals m_1 and m_2 . A lune of dihedral angle β has area 2β , so $\mathcal{G}$ has area $2(\pi - \alpha)$, and dividing by the surface 4π of the unit sphere gives $\mu = (\pi - \alpha)/(2\pi)$. Explicitly,

$$\alpha = \arccos \left(\frac{\lambda(\varphi + \sigma)\sigma_d^2 - \omega\eta\sigma_a^2}{\sqrt{\lambda^2(\varphi + \sigma)^2\sigma_d^2 + \omega^2\sigma_a^2 + \sigma_s^2} \sqrt{\sigma_d^2 + \eta^2\sigma_a^2}} \right),$$

since $m_1' m_2 = \lambda(\varphi + \sigma)\sigma_d^2 - \omega\eta\sigma_a^2$, $\|m_1\| = \sqrt{\lambda^2(\varphi + \sigma)^2\sigma_d^2 + \omega^2\sigma_a^2 + \sigma_s^2}$ and $\|m_2\| = \sqrt{\sigma_d^2 + \eta^2\sigma_a^2}$.

Step 4. Taking the ratio of Steps 2 and 3 gives $\gamma = \delta/\mu = \arctan(\sigma_s/(\omega\sigma_a))/(\pi - \alpha)$.

■

C REPLICABILITY OF SHOCK SYSTEMS

This appendix extends the replicability measures of Section 2 to settings where sign restrictions are imposed on $k > 1$ shocks, and describes the algorithm used to sample the associated conditional measures. Throughout, the target remains the first shock, so the zero-weight constraint applies to the first column of the rotation only; this is the object computed in Sections 4 and 5, where Assumptions 2 and 3 restrict two shocks while the shock of interest is one.

Single-shock replicability with multiple-shock restrictions ($k > 1$). Suppose we target $\epsilon_{1,t}$ but impose sign restrictions on the first $k > 1$ shocks. For each $j = 1, \dots, k$ define

$$M^{(j)} \equiv \text{diag}(\text{sign}(\tilde{f}_j)) \tilde{F}.$$

Replicability requires existence of $Q = [q_1 \cdots q_k] \in \mathbb{R}^{n \times k}$ such that

$$M^{(j)} q_j > 0 \quad \forall j = 1, \dots, k, \quad Q' Q = I_k, \quad e_1' q_1 = 0.$$

Let $\mathcal{K}^{(j)} = \{q \in \mathbb{R}^n : M^{(j)}q > 0\}$ and, as in Section 2, $\mathcal{H} = \{q \in \mathbb{R}^n : e'_1 q = 0\}$. The feasible set is

$$\mathcal{F}_{1,k} = \left\{ Q \in \text{St}(n, k) \mid q_1 \in \mathcal{K}^{(1)} \cap \mathcal{H}, q_j \in \mathcal{K}^{(j)} \forall j = 2, \dots, k \right\},$$

where $\text{St}(n, k) = \{Q \in \mathbb{R}^{n \times k} : Q'Q = I_k\}$ is the Stiefel manifold. As in the single-shock case, the relevant measures are conditional masses. Define

$$\delta_{1,k} \equiv \Pr\left(M^{(j)}q_j > 0 \forall j \mid Q \in \text{St}(n, k), e'_1 q_1 = 0\right), \quad \mu_k \equiv \Pr\left(M^{(j)}q_j > 0 \forall j \mid Q \in \text{St}(n, k)\right),$$

where Q is drawn from the uniform (Haar) distribution on the Stiefel manifold, in the first case conditional on the zero-weight constraint $e'_1 q_1 = 0$ (see the algorithm below). The corresponding replica ratio is

$$\gamma_{1,k} \equiv \frac{\delta_{1,k}}{\mu_k},$$

with the same interpretation as in the single-shock case: the relative likelihood that the restriction system is satisfied when the target shock is excluded from the first column of the rotation.

System replicability for k shocks. Extending to a system of the first k shocks imposes no-self-use constraints for each:

$$M^{(j)}q_j > 0 \quad \forall j = 1, \dots, k, \quad Q'Q = I_k, \quad e'_j q_j = 0 \quad \forall j = 1, \dots, k.$$

Let $H^{(j)} = \{q : e'_j q = 0\}$. The feasible set is

$$\mathcal{F}_{k,k} = \left\{ Q \in \text{St}(n, k) \mid q_j \in \mathcal{K}^{(j)} \cap H^{(j)} \forall j = 1, \dots, k \right\},$$

and the corresponding mass is

$$\delta_{k,k} \equiv \Pr\left(M^{(j)}q_j > 0 \forall j \mid Q \in \text{St}(n, k), e'_j q_j = 0 \forall j\right),$$

where Q is drawn from the Haar distribution conditional on the zero-diagonal constraints, with associated replica ratio $\gamma_{k,k} \equiv \delta_{k,k}/\mu_k$. Single-shock and system replicability need not factorize: in general $\delta_{k,k} \neq \prod_{j=1}^k \delta_j$. When cones are large (replication is easy), orthogonality constraints become binding, so system mass can be much smaller than the product of marginal masses.

Sampling the conditional measures. Both $\delta_{1,k}$ and μ_k are computed by acceptance sampling from the same routine, which differs only in whether the zero-weight constraint is imposed on the first column. Let $c \in \{0, 1\}$ index the two cases, with $c = 1$ delivering the constrained draw used for $\delta_{1,k}$ and $c = 0$ the unconstrained draw used for μ_k . Write $Q_{j-1} = [q_1 \cdots q_{j-1}]$.

1. Draw $z \sim \mathcal{N}(0, I_n)$. If $c = 1$, replace z by $(I_n - e_1 e_1')z$, i.e. set its first element to zero. Set $q_1 = z/\|z\|$.
2. For $j = 2, \dots, k$: draw $z \sim \mathcal{N}(0, I_n)$, project it onto the orthogonal complement of the columns already accepted,

$$\tilde{z} = z - Q_{j-1} (Q_{j-1}' z),$$

and set $q_j = \tilde{z}/\|\tilde{z}\|$ (redraw in the numerically degenerate case $\tilde{z} \approx 0$).

3. Collect $Q = [q_1 \cdots q_k]$ and accept the draw if $M^{(j)} q_j > 0$ for all $j = 1, \dots, k$.

Step 1 with $c = 1$ draws q_1 uniformly on the unit sphere of $\mathcal{H}$, a copy of $\mathbb{S}^{n-2}$; with $c = 0$ it draws q_1 uniformly on $\mathbb{S}^{n-1}$. Step 2 completes the frame by Gram–Schmidt orthogonalization of independent Gaussian vectors, which delivers the Haar measure on the Stiefel manifold of $q_1^\perp$ and imposes no restriction on the remaining columns beyond orthonormality. The resulting Q is therefore distributed as Haar on $\text{St}(n, k)$, conditional on $e_1' q_1 = 0$ when $c = 1$ and unconditionally when $c = 0$. Note in particular that no constraint of the form $e_j' q_j = 0$ is imposed for $j \geq 2$: the target is the first shock alone, and $\delta_{1,k}$ conditions on a single zero on the diagonal of the rotation, not on all k of them.

Repeating the routine for $\ell = 1, \dots, L$ and recording acceptance frequencies gives

$$\hat{\delta}_{1,k} = \frac{1}{L} \sum_{\ell=1}^L \mathbf{1} \left\{ M^{(j)} q_j^{(\ell)} > 0 \ \forall j \right\}_{c=1}, \quad \hat{\mu}_k = \frac{1}{L} \sum_{\ell=1}^L \mathbf{1} \left\{ M^{(j)} q_j^{(\ell)} > 0 \ \forall j \right\}_{c=0}, \quad \hat{\gamma}_{1,k} = \frac{\hat{\delta}_{1,k}}{\hat{\mu}_k}.$$

Because the two masses are produced by the same code path, $\hat{\gamma}_{1,k}$ is a ratio of consistently constructed quantities.

Sign normalization. The acceptance step above tests $M^{(j)} q_j > 0$ for the drawn columns, without allowing the sign flips $q_j \mapsto -q_j$ that are sometimes used to enlarge the accepted set (Rubio-Ramírez et al., 2010). Since the events $\{M^{(j)} q_j > 0\}$ and $\{M^{(j)}(-q_j) > 0\}$ are disjoint and the Haar measure is invariant under $q_j \mapsto -q_j$, admitting flips would multiply both $\delta_{1,k}$ and μ_k by exactly 2^k and leave $\gamma_{1,k}$ unchanged.

We report the unflipped masses throughout, consistently with the analytical calculations of Section 3.

D MONETARY POLICY SHOCK REPLICABILITY

As discussed in Wolf (2020), combinations of supply and demand shocks can reproduce the sign pattern of inflation and the interest rate generated by a monetary policy shock—a phenomenon referred to as *masquerading*. However, while a contractionary monetary policy shock lowers output on impact, the masquerading combination of a technology shock (η^a) and a preference shock (η^b) raises output on impact. As a result, agnostic sign restrictions that only target the comovement of inflation and the interest rate (e.g. Uhlig, 2005) are not sufficient to recover the transmission of monetary policy to output. The goal of this section is to revisit this issue and provide probability statements about monetary policy shock replicability, the likelihood of selecting a replica, and the implied transmission.

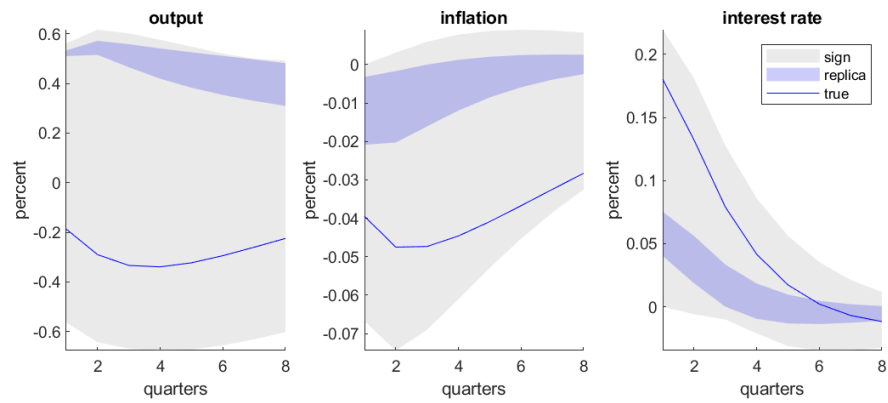
Following Wolf (2020), we assume that the DGP features three shocks: preference (η^b), technology (η^a), and monetary policy (η^m). The econometrician observes output, inflation, and the policy rate, and seeks to identify monetary policy using an agnostic sign scheme:

Assumption 4 *A monetary policy shock moves the policy rate and inflation in opposite directions on impact and for the subsequent two quarters.*

Under Assumption 4, the monetary policy shock is replicable by a linear combination of the two non-policy shocks, and the replicability mass is 10%. The replica ratio—the probability that Assumption 4 is satisfied by a combination placing zero weight on the true policy shock, relative to the probability that it is satisfied by an unrestricted combination—is 0.25. Thus, the restrictions reduce the prevalence of (near-)replicas of the policy shock only by a factor of four relative to the uniform benchmark (Remark 3).

Figure 12 illustrates the transmission of the monetary policy shock and its replica. The replica reproduces the intended inflation–interest-rate pattern (inflation falls as the policy rate rises) but moves output in the opposite direction relative to a contractionary policy shock.

Figure 12: Dynamic transmission of the monetary policy shock and a non-policy replica.



Notes: The figure shows the identified set of impulse responses to a monetary policy shock under Assumption 4 (gray) and the impulse responses of a replica constructed from non-policy shocks (blue).